

Embassies and Democracy - Results Only

Steven Pappas and Gregory J. Matthews

2026-04-27

Summary

BROAD QUESTION

Broad question: Does the process of democratization within African countries affect the formulation and implementation of African foreign policies.

THERE ARE THREE DIFFERENT SETS OF STATISTICALLY BASED QUESTIONS (We address only question 1 here):

1. African democracy and African embassies. Do greater levels of democratization of African countries lead to larger and more diversified embassy networks maintained by African countries overseas? Dependent variable: embassies maintained abroad (yes/no by country) in 1965, 1975, 1985, 1995, 2000, 2005, 2010, 2015, 2020, 2025 (use 2024 for 2025).
2. African democracy and African trade. Do greater levels of democratization within African countries lead to larger and more diversified levels of trade imports and trade exports by African countries? Dependent variable: \$ amount of imports and exports annually for 1960 to 2022)
3. African democracy and aid to Africa. Do greater levels of democratization within African countries lead to larger and more diversified levels of foreign aid to African countries? (Dependent variable:\$ amount of aid annually for every country in the world 1960 to 2021).

THUS, THREE DIFFERENT DEPENDENT VARIABLES:

1. Total number and diversification of African embassies maintained abroad.
2. Total amount and diversification of African trade imports and trade exports
3. Total amount and diversification of foreign aid received by African countries

THREE DIFFERENT VARIABLES OF AFRICAN DEMOCRACY AS INDEPENDENT VARIABLE

Three different variables of democracy (each with different inclusive dates) to use separately in the analyses as the independent variable:

1. V-Dem democracy variable (1951-2024) Variable name in the dataset: v2x_LIBDEM. It ranges from 0 (worst) to 1 (best).
2. Polity IV democracy variable (1948-2018) Variable name in the dataset: POLITY. It ranges from -10 (worst) to +10 (best).
3. Freedom House democracy variable (1972 - 2024) Variable name in the dataset: TOTLIB. It ranges from 1 (best) to 14 (worst). It is a combination of the following two variables, which people sometimes run separately: Civil liberties - Variable name in the dataset: -CIVLIB. It ranges from 1 (best) to 7 (worst). -Political rights: Variable name in the dataset: POLLIB. It ranges from 1 (best) to 7 (worst).

For each of the three democracy variables, will do a series of 4 tests with lags. -Test 1. Change in democracy score (no lag) -Test 2. Change in democracy score (1 year lag) -Test 3. Change in democracy score (2 year lag) -Test 4. Change in democracy score (3 year lag)

CONTROL VARIABLES RELATED FOR AFRICAN COUNTRIES (AFRICAN FACTORS)

-African GNI OR GNI per capita (logged?) -African Population (size in millions) - Region of Africa (1= North Africa, 2 = East Africa, 3 = Southern Africa, 4= Central Africa, 5 = West Africa) - African Colonial Past (Colpast2) (1 = French, 2 = British, 3 = Portuguese, 4 = Spanish, 5 = Italian, 6 = Belgian, 7 = German, 8 = United States, 9 = Not colony, 10 = unique) - African Ideology (1 = capitalist, 2 = socialist, 3 = Marxist) - African Historical Period - Entire independence era - Cold War era (1945-1989) - Post-Cold War era (1990-present)

PULL FACTORS—CHARACTERISTICS OF OTHER NON-AFRICAN COUNTRIES OF THE WORLD—JUST FOR THE EMBASSY QUESTION

- Level of democracy (Freedom House)
- Level of democracy (Polity IV)
- Level of democracy (V-dem) These are in the data set for the African countries, but need to be added for the rest of the world.
- GNI
- GNI per capita
- Population
- Imports
- Exports

- Foreign Aid

FOR THE THREE SETS OF STATISTICALLY BASED QUESTIONS, LET'S START WITH THE EMBASSIES (ABOVE QUESTION #1 FIRST)

ANSWERING QUESTION #1

To evaluate this question we first created three sets of reduced and full models. Our reduced models sought to evaluate the relationship and the significance of the relationship between a country's embassy count and its democracy score from Freedom House, Polity IV, and V-dem. In these models, democracy score was the only explanatory variable after controlling for the country and year with random effects. The model equation is given below, where γ_{0k} represents the random effect for country k and γ_{0t} is the random effect for year t . We follow the same setup for the lagged models, which go back as far as three years for each democracy score.

Number of embassy models

Reduced model

$$Y_{kt} \sim \text{Poisson}(\lambda_{kt})$$

$$\log(\lambda_{kt}) = \beta_0 + \beta_1 X_{1kt} + \gamma_{0k} + \gamma_{0t} + \epsilon_{kt}$$

$$\gamma_{0k} \sim N(0, \sigma_k^2)$$

$$\gamma_{0t} \sim N(0, \sigma_t^2)$$

where Y_{kt} is the number of embassies for country k in year t and X_{1kt} is one of the democracy scores for country k in year t . We also considered models where the democracy score as a predictor was lagged by 1, 2, or 3 years (i.e. for a single lag: $\log(\lambda_{kt}) = \beta_0 + \beta_1 X_{1k(t-1)} + \gamma_{0k} + \gamma_{0t} + \epsilon_{kt}$ where $X_{1k(t-1)}$ is the democracy score in the prio year.)

As shown in Table 1, in these reduced models none of the democracy scores or their lags display significance at an $\alpha = 0.05$ confidence level.

Full model

The full models (results in Table 2) follow a similar setup as the reduced models, but they include additional explanatory variables for the country's GNI, population, region, colonization status after World War II, ideology, and whether the year is during the Cold War. We applied log transformations to GNI and population, since they displayed a right-skewed distribution. We also kept the reduced models' random effects for country and year and followed a similar setup for lagged democracy scores, going back as far as three years. The model equation is given below, with k and t again representing country and year, respectively.

$$Y_{kt} \sim \text{Poisson}(\lambda_{kt})$$

$$\log(\lambda_{kt}) = \beta_0 + \beta_1 X_{1kt} + \dots + \beta_7 X_{7kt} + \gamma_{0k} + \gamma_{0t} + \epsilon_{kt}$$

$$\gamma_{0k} \sim N(0, \sigma_k^2)$$

$$\gamma_{0t} \sim N(0, \sigma_t^2)$$

where again Y_{kt} is the number of embassies for country k in year t and X_{1kt} is one of the democracy scores for country k in year t . The additional X 's represent country's GNI, population, region, colonization status after World War II, ideology, and whether the year is during the Cold War. We also considered models where the democracy score as a predictor was lagged by 1, 2, or 3 years (i.e. for a single lag: $\log(\lambda_{kt}) = \beta_0 + \beta_1 X_{1k(t-1)} + \gamma_{0k} + \gamma_{0t} + \epsilon_{kt}$ where $X_{1k(t-1)}$ is the democracy score in the prior year.)

Similar to the reduced models, in the full models none of the democracy scores or their lags display significance at an $\alpha = 0.05$ confidence level. Additionally, none reached a p-value under 0.2. Full results are shown in Table 2.

Change in number of embassy models

Reduced model

Next, we sought to evaluate the relationship between a country's yearly change in embassy count by the change in their democracy score. Once again, in this first set of models the random effects for country and year were included. The model equations follow the previously described reduced model setup, however the response variable here is the change in the number

of embassies from one year to the next as opposed to the total number of embassies. Therefore, a normal model was used instead of a Poisson model. The details of the model are below:

$$\Delta_{kt} \sim \text{Normal}(\mu_{kt}, \sigma^2)$$

$$\log(\mu_{kt}) = \beta_0 + \beta_1 X_{1kt} + \gamma_{0k} + \gamma_{0t} + \epsilon_{kt}$$

$$\gamma_{0k} \sim N(0, \sigma_k^2)$$

$$\gamma_{0t} \sim N(0, \sigma_t^2)$$

where Δ_{kt} is the change in the number of embassies from year $t - 1$ to t , while γ_{0k} , γ_{0t} , and X_{1kt} are the same as before.

The tables show no significance for any of the changes in democracy score at an $\alpha = 0.05$ confidence level. The closest is the unlagged democracy score from Polity IV, with a p-value of 0.140. Full results are shown in table 3.

Full Model

Next, we sought to evaluate the relationship between a country's yearly change in embassy count by the change in their democracy score while accounting for the variables from our previous full model and random effects for country and year. It is important to note that some models only had realized values for before or after the Cold War, so we removed the Cold War indicator in models where it could not explain any variation in our response variable. We also applied the same log transformations to GNI and population to reduce the effect of their right-skew.

The tables show no significance for any of the changes in democracy score at an $\alpha = 0.05$ confidence level, even when accounting for the full range of factors. The closest is the second lag for the democracy score from Freedom House, with a p-value of 0.102. Full results are shown in Table 4.

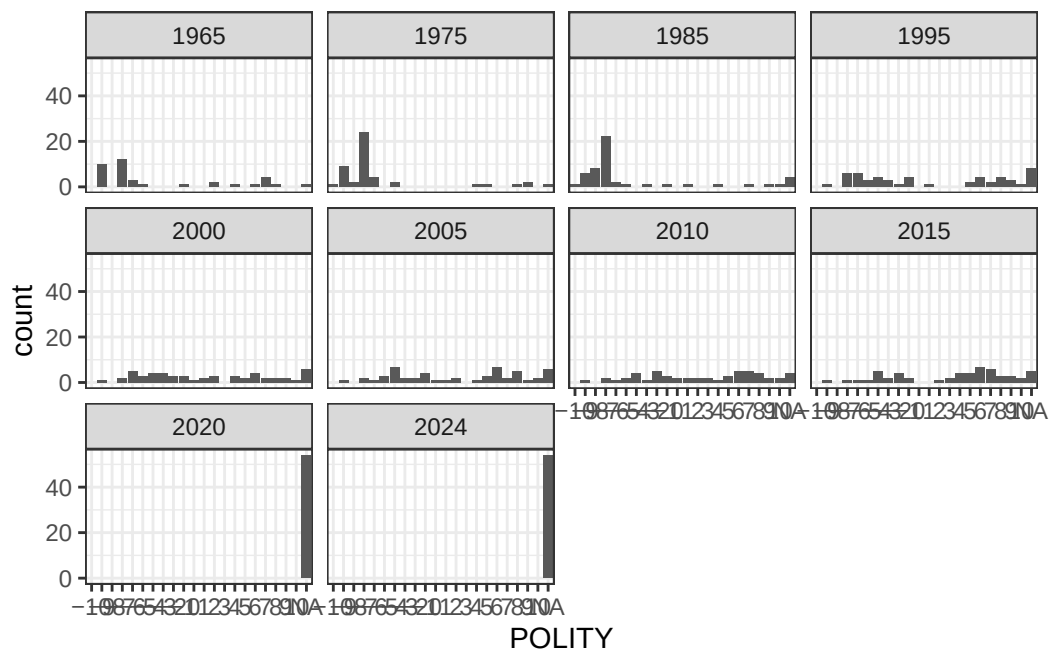
Full model with differenced predictors

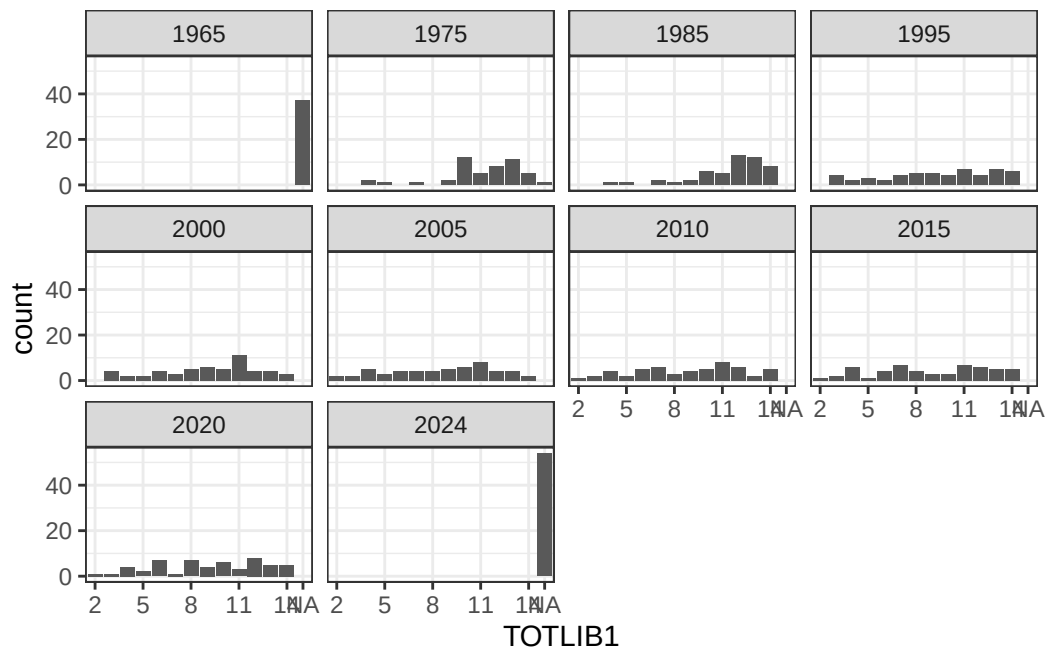
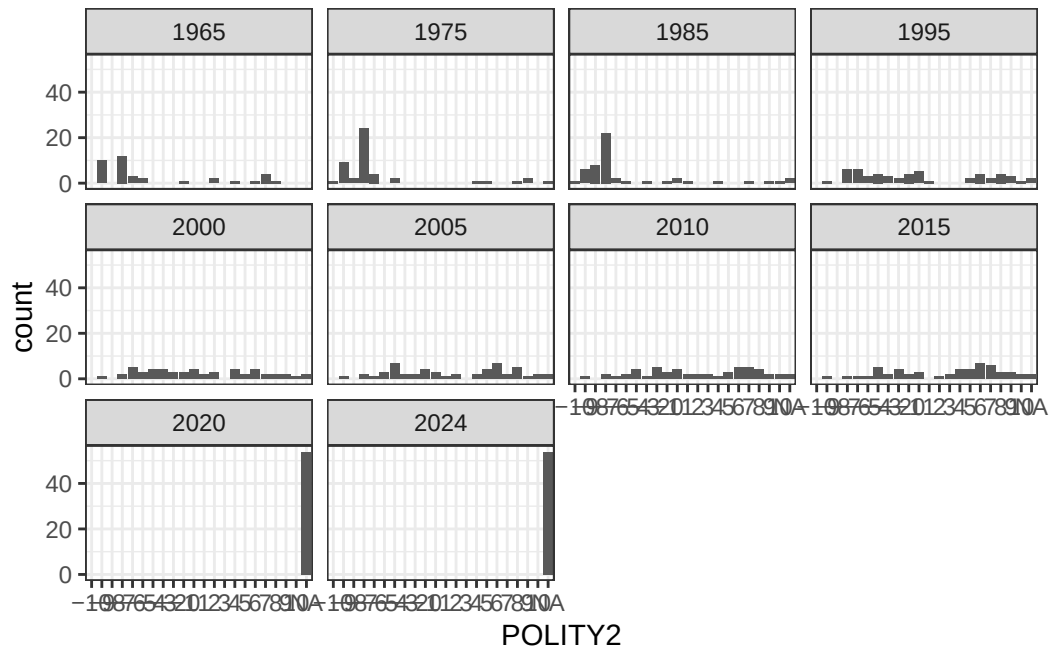
Finally, we replicated the setup from our previous full model but applied the same yearly change to the log-transformed GNI and population variables in addition to the democracy score. However, we only lagged the democracy score, so every other variable matched the year the change in embassy was measured. We kept the same random variables for country and year.

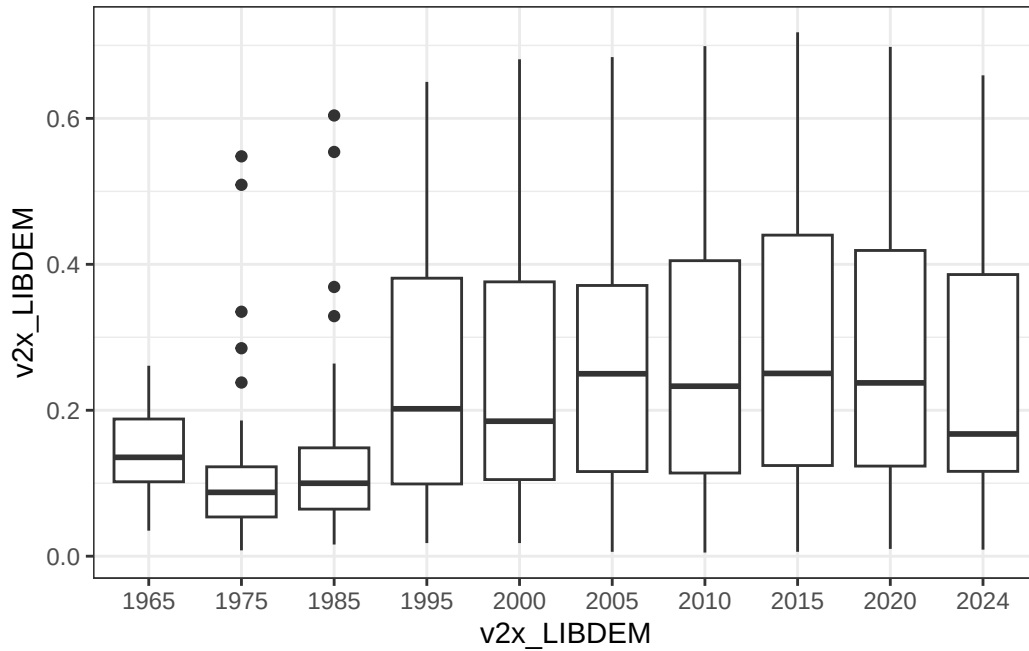
Similar to the previous full model, the tables show no significance for any of the changes in democracy score at an $\alpha = 0.05$ confidence level. The closest is the nonlagged democracy score from V-Dem, with a p-value of 0.171. Full results of these models are shown in Table 5.

Exploratory Data Analysis

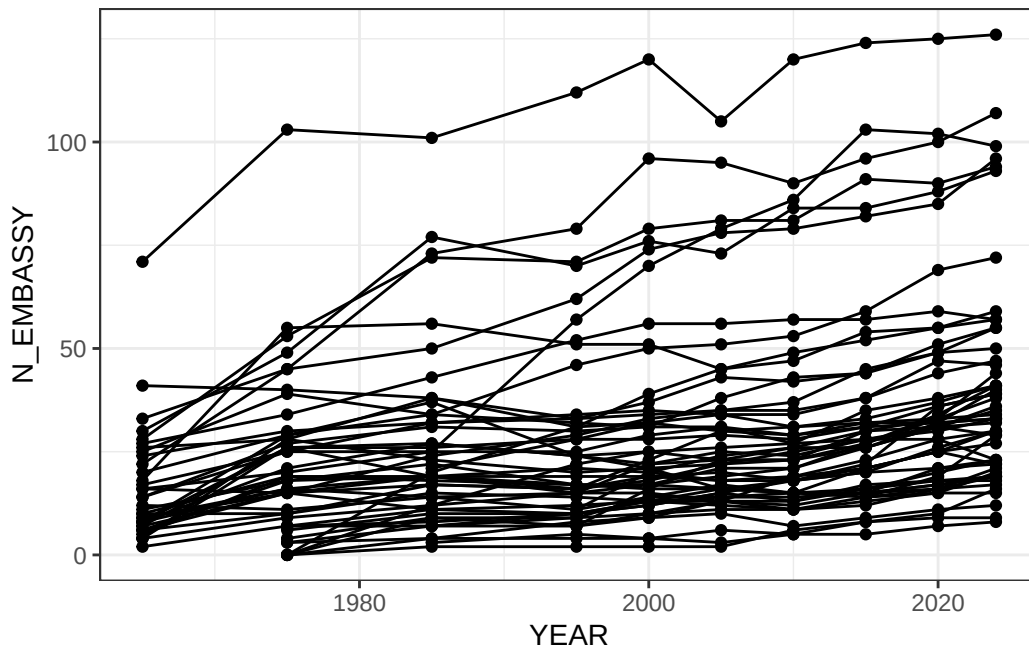
Below are the distributions of the three democracy variables.



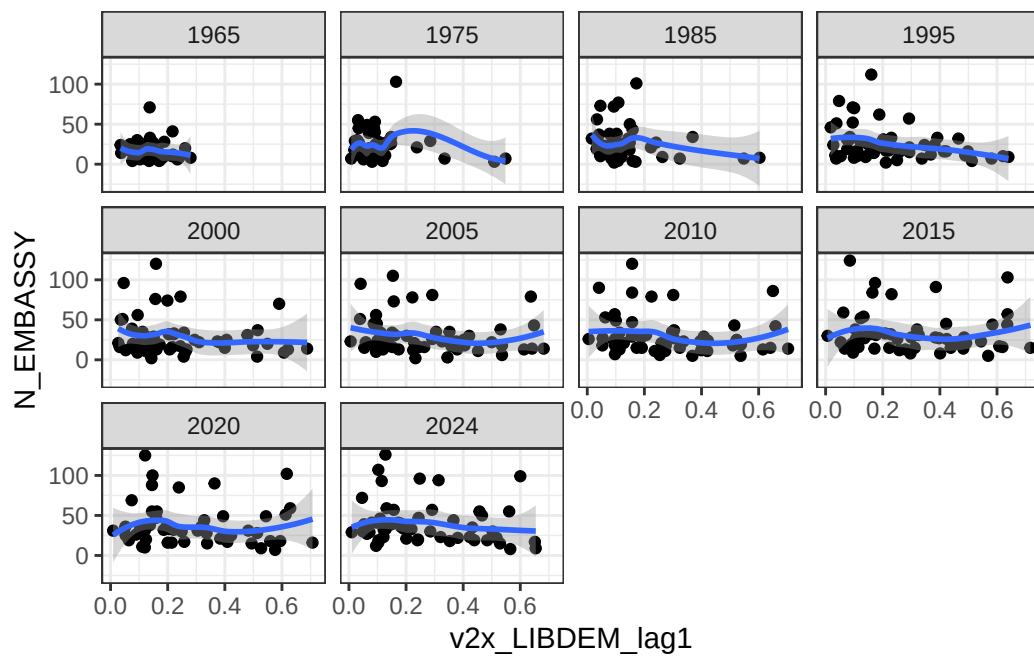
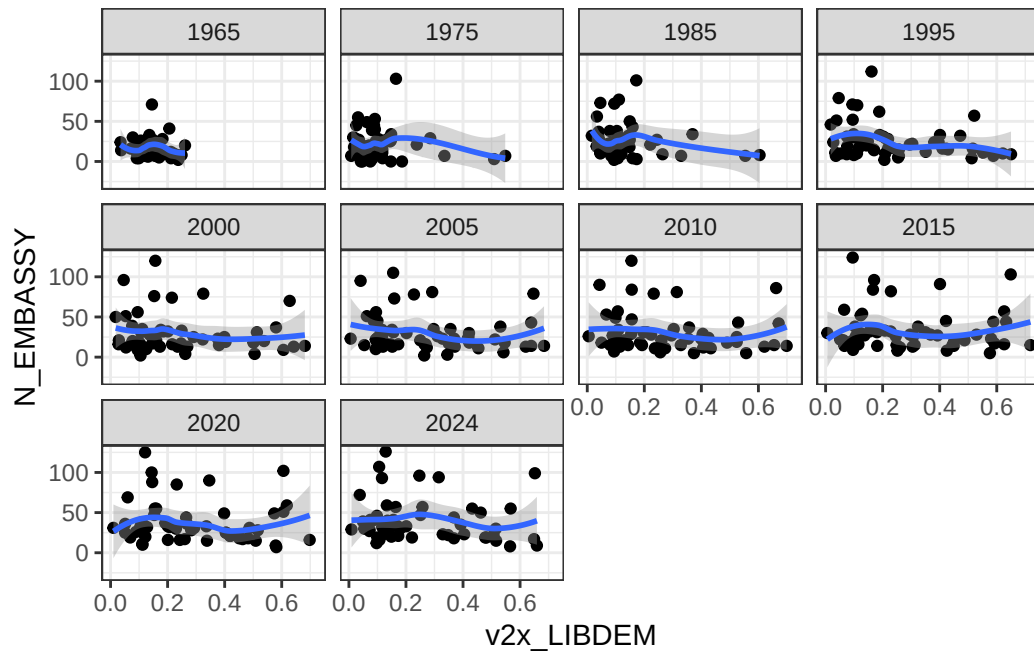


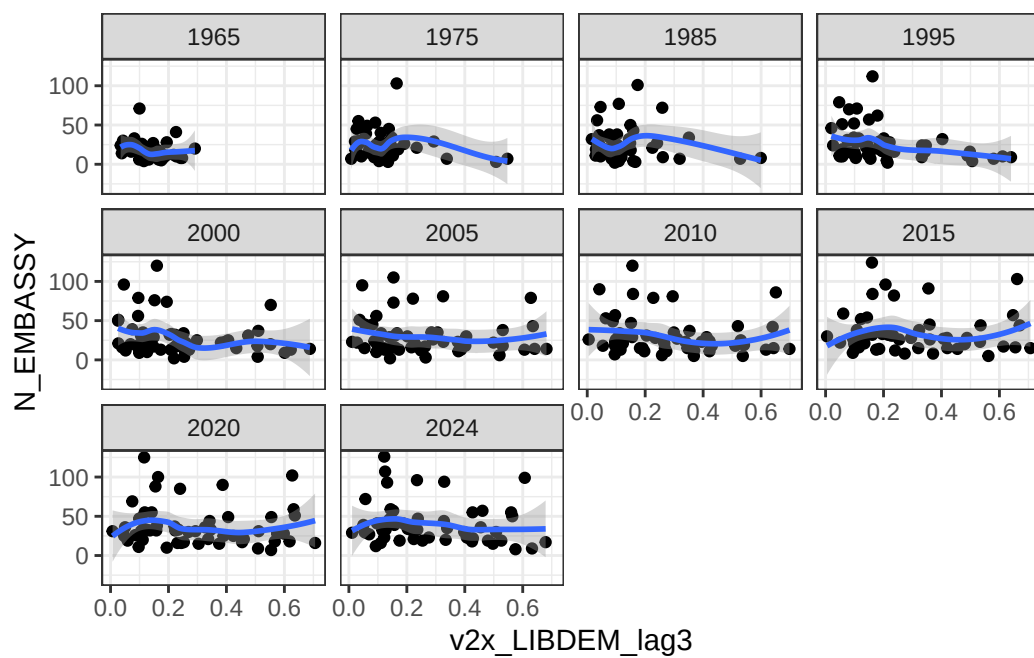
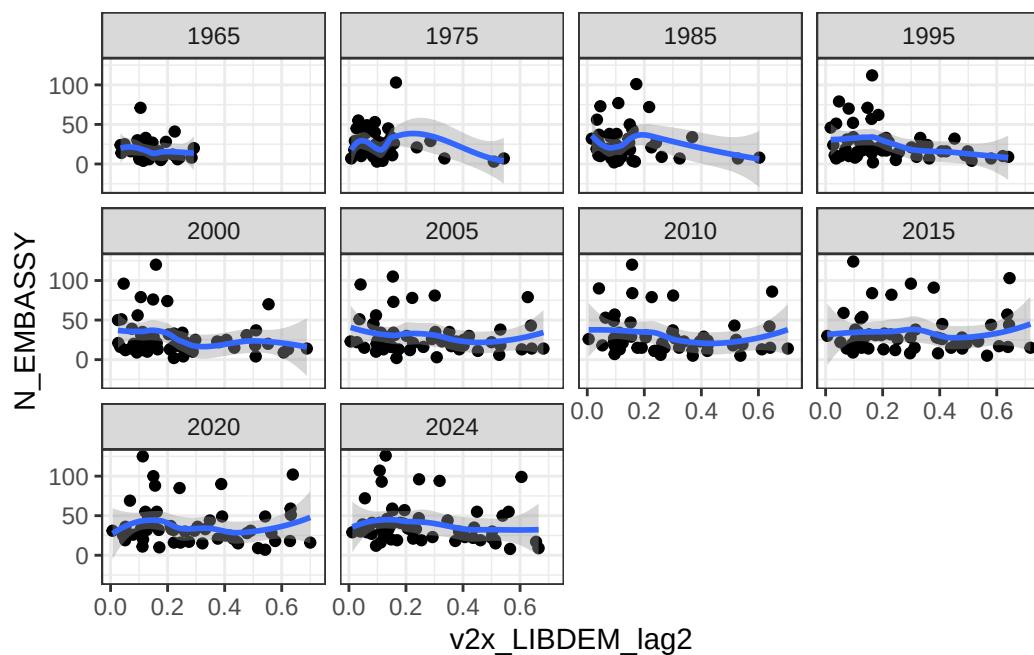


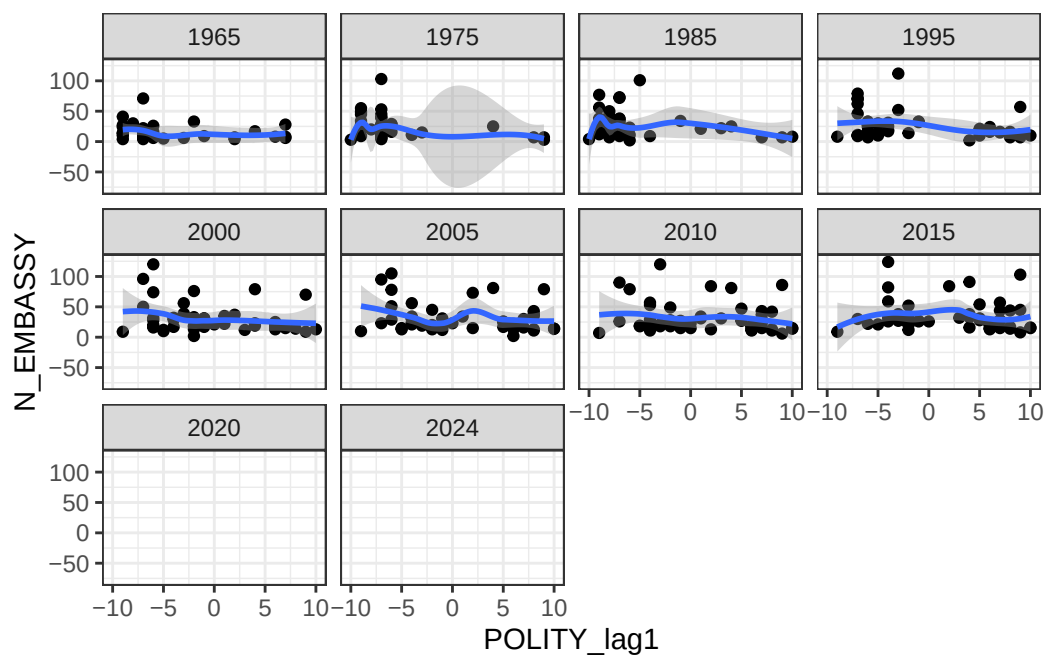
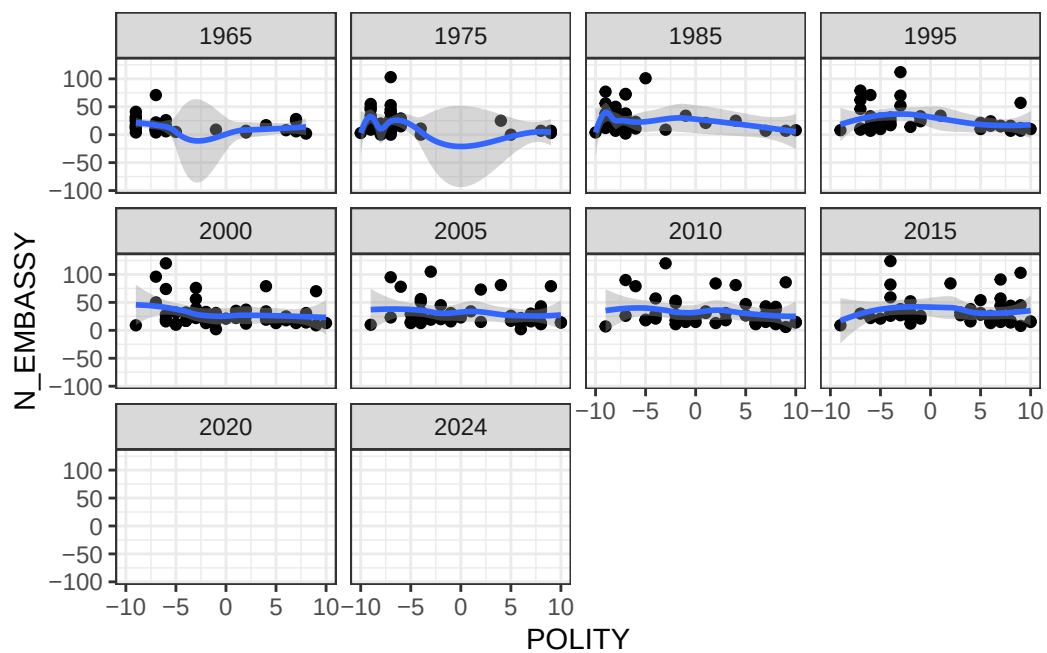
The plot below is the number of embassies over time with each connected line representing a single country.

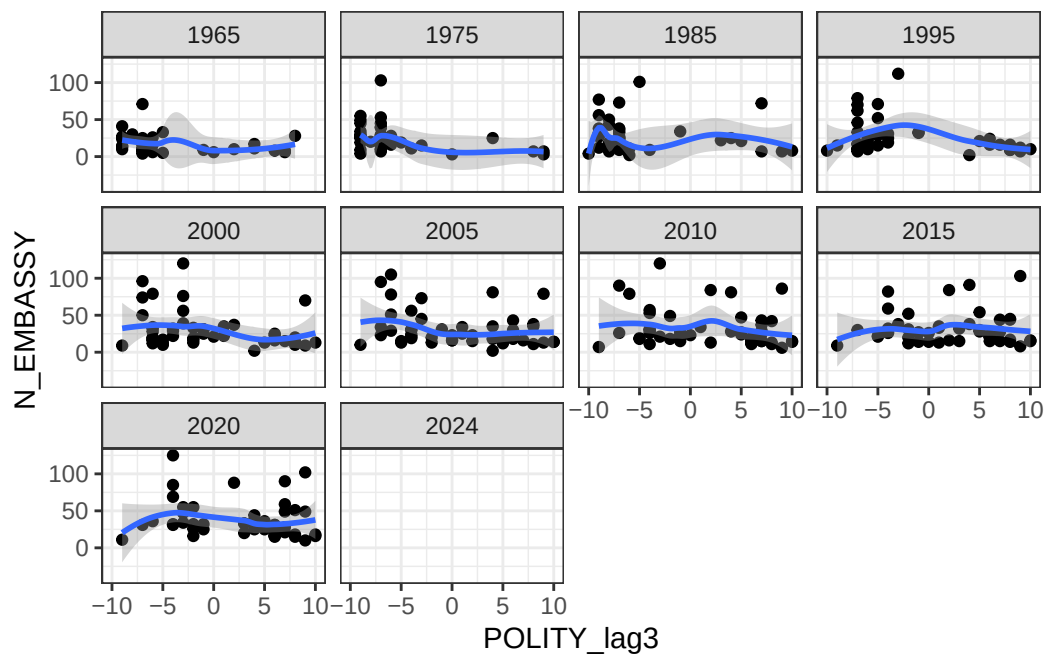
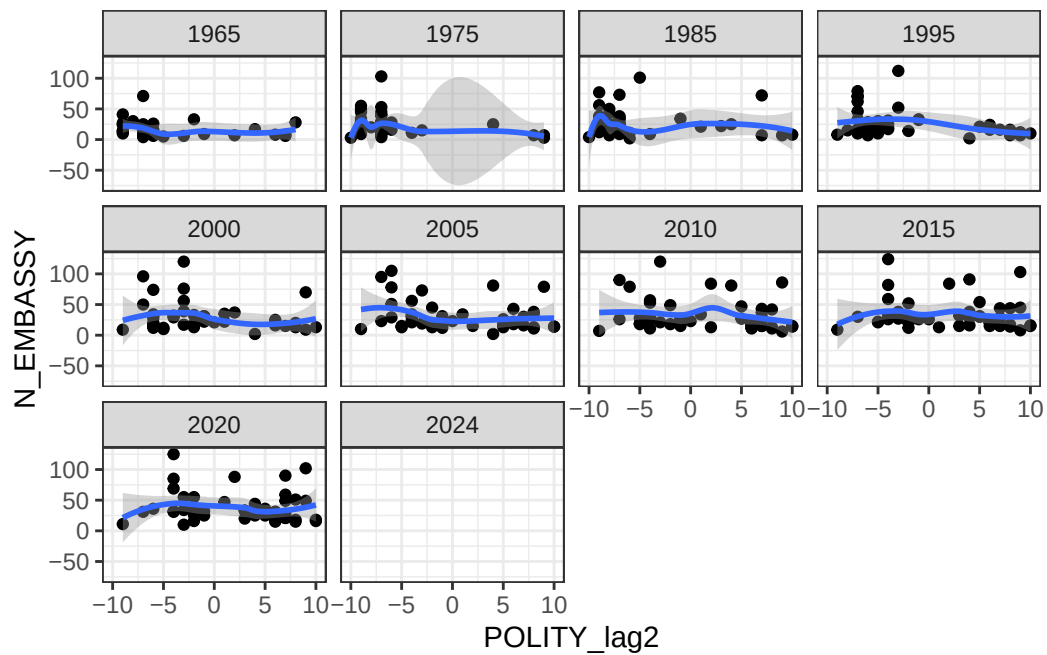


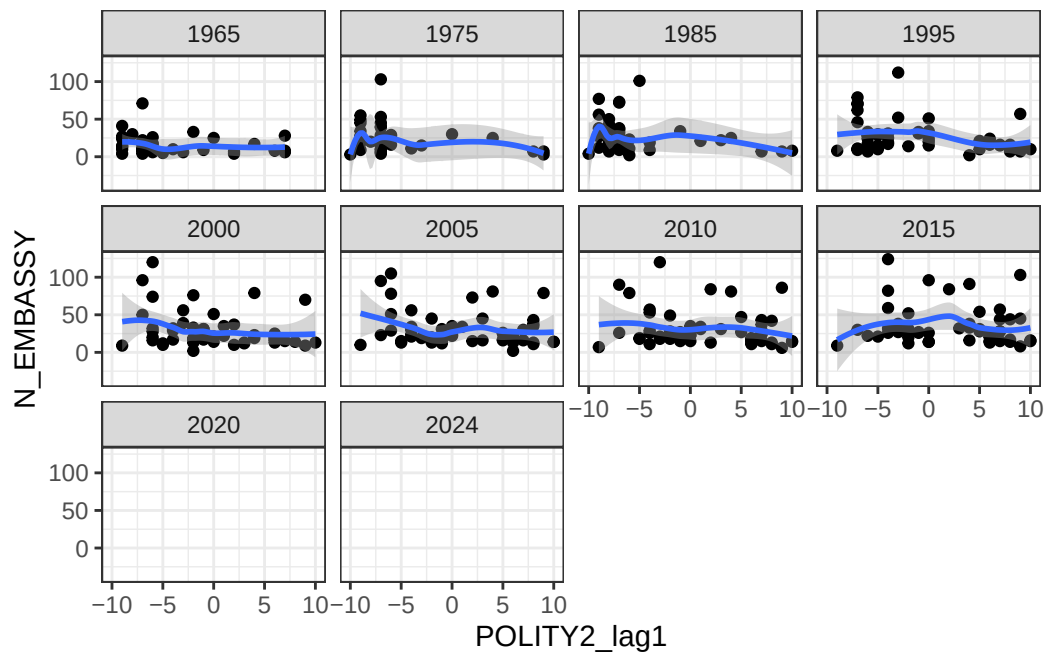
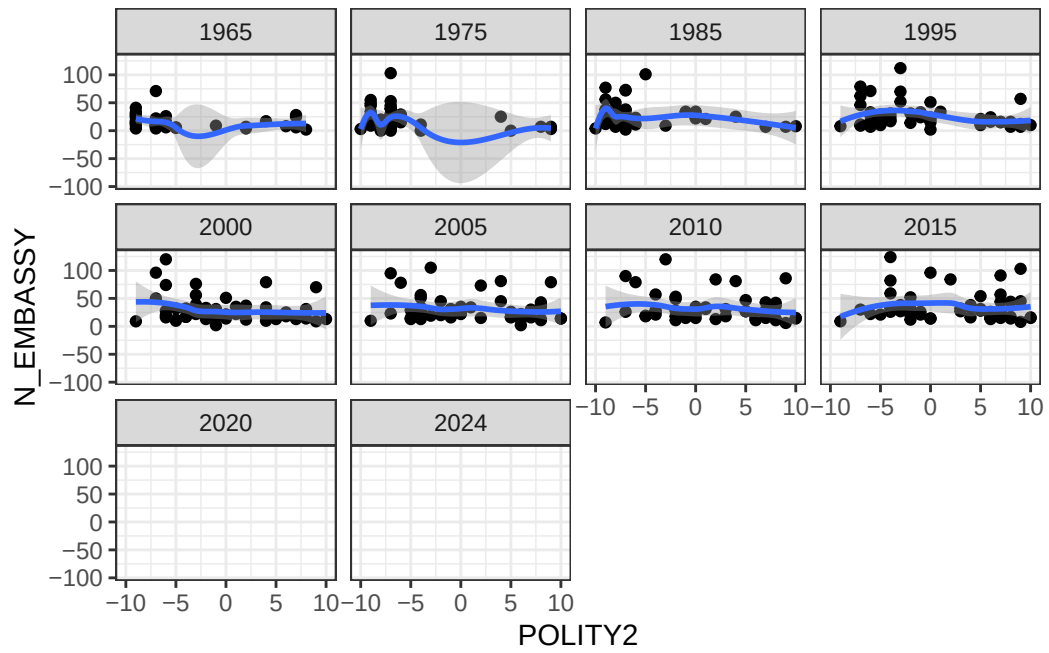
Next, scatter plots of the number of the democracy variables versus the embassies are plotted and each plot includes a trend line of the relationship.

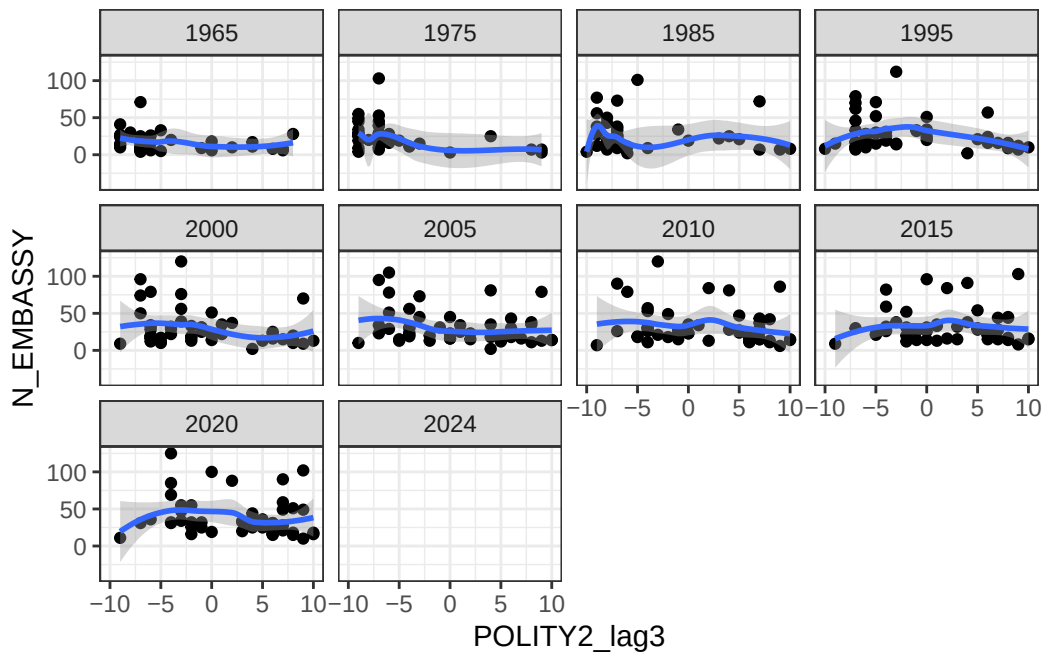
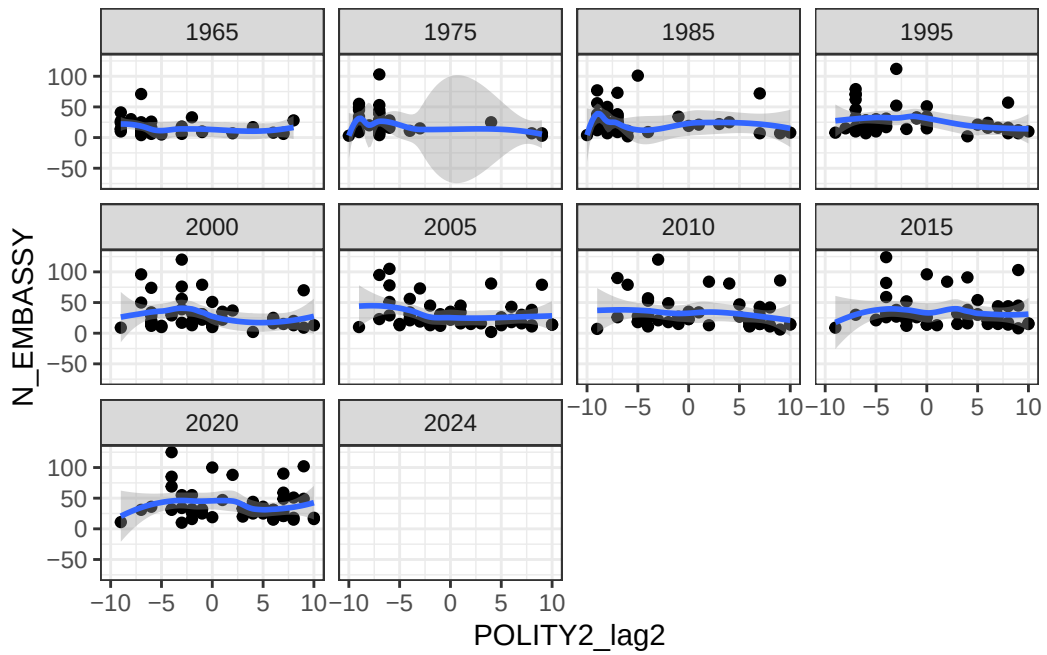


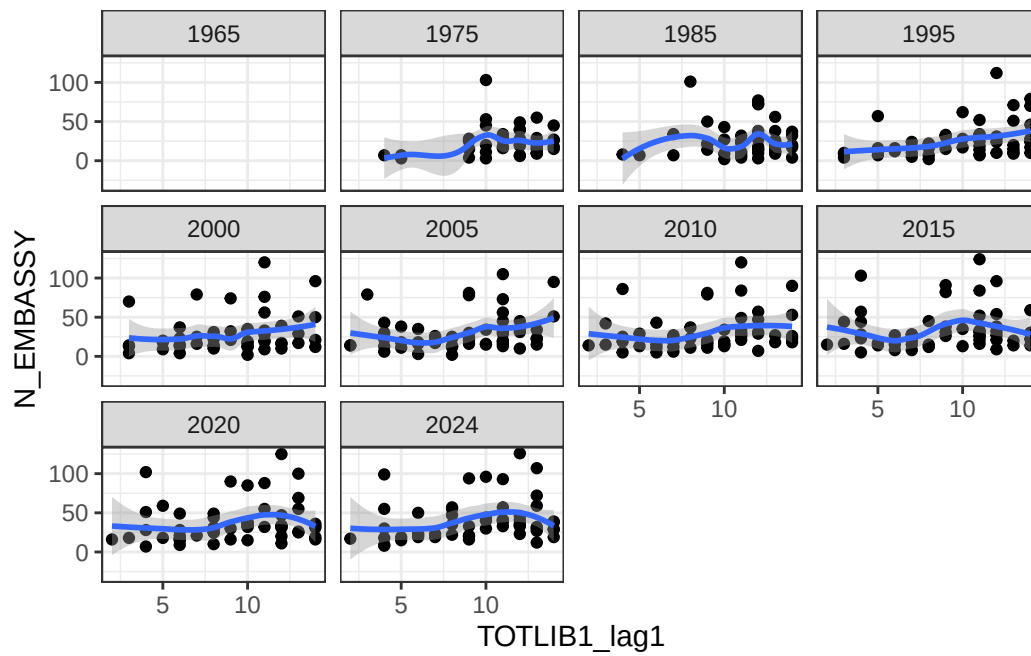
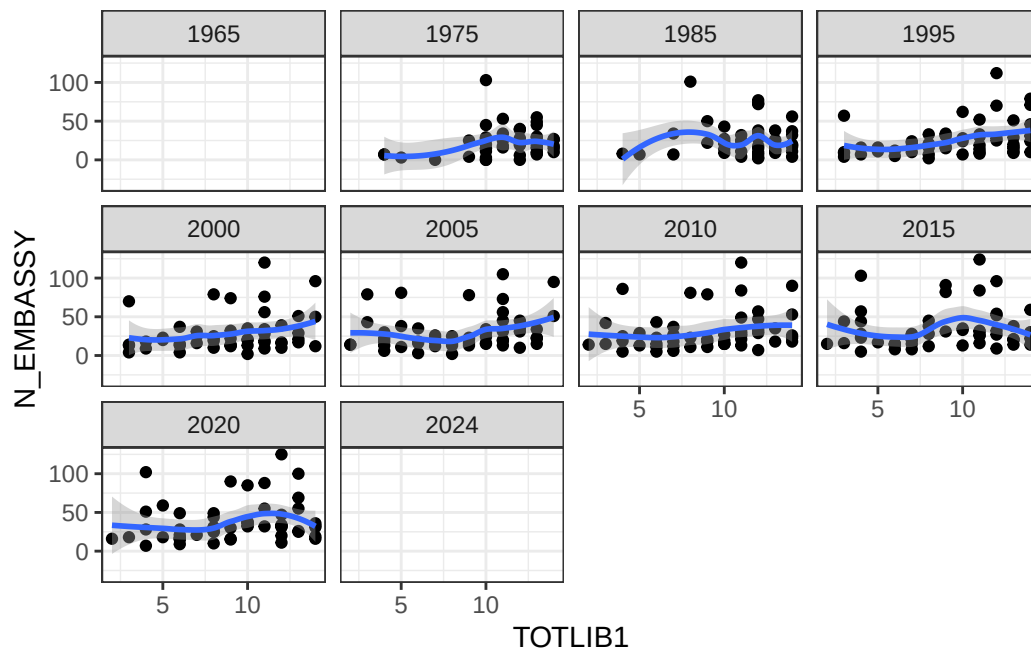


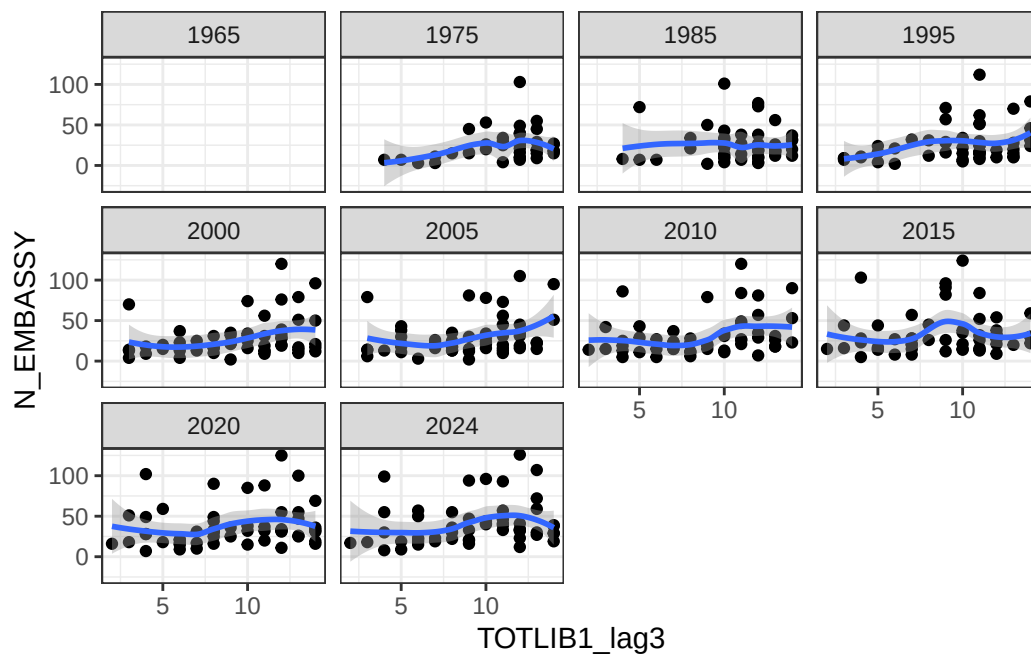
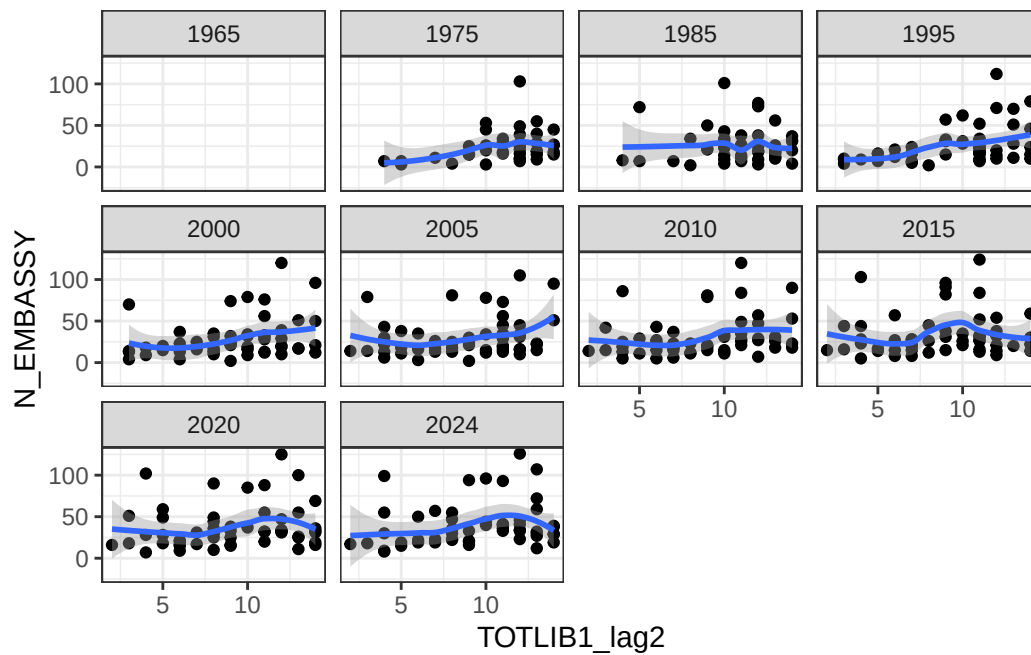












```
#Correlations
#Polity
modelset %>% filter(YEAR < 2020) %>% group_by(YEAR) %>%
  summarize(cor = cor(N_EMBASSY,POLITY, use = "pairwise.complete.obs") , pval = cor.test(N_E
```



```
# A tibble: 8 x 3
  YEAR      cor  pval
  <dbl>   <dbl> <dbl>
1  1965 -0.227  0.182
2  1975 -0.305  0.0370
3  1985 -0.211  0.155
4  1995 -0.290  0.0530
5  2000 -0.247  0.0943
6  2005 -0.211  0.155
7  2010 -0.200  0.168
8  2015 -0.0811 0.580
```

```
modelset %>% filter(YEAR < 2020) %>% group_by(YEAR) %>%
  summarize(cor = cor(N_EMBASSY,POLITY_lag1, use = "pairwise.complete.obs") , pval = cor.test
```

```
# A tibble: 8 x 3
  YEAR      cor  pval
  <dbl>   <dbl> <dbl>
1  1965 -0.192  0.285
2  1975 -0.288  0.0640
3  1985 -0.218  0.137
4  1995 -0.330  0.0269
5  2000 -0.242  0.109
6  2005 -0.218  0.146
7  2010 -0.182  0.210
8  2015 -0.111  0.448
```

```
modelset %>% filter(YEAR < 2020) %>% group_by(YEAR) %>%
  summarize(cor = cor(N_EMBASSY,POLITY_lag2, use = "pairwise.complete.obs") , pval = cor.test
```

```
# A tibble: 8 x 3
  YEAR      cor  pval
  <dbl>   <dbl> <dbl>
1  1965 -0.194  0.287
2  1975 -0.302  0.0523
3  1985 -0.0749 0.613
4  1995 -0.372  0.0129
5  2000 -0.225  0.137
6  2005 -0.244  0.102
7  2010 -0.208  0.152
8  2015 -0.105  0.478
```

```
modelset %>% filter(YEAR < 2020) %>% group_by(YEAR) %>%
  summarize(cor = cor(N_EMBASSY,POLITY_lag3, use = "pairwise.complete.obs") , pval = cor.test
```

```
# A tibble: 8 x 3
  YEAR      cor    pval
<dbl> <dbl> <dbl>
1  1965 -0.241  0.192
2  1975 -0.350  0.0232
3  1985 -0.0776 0.600
4  1995 -0.310  0.0461
5  2000 -0.281  0.0529
6  2005 -0.266  0.0647
7  2010 -0.215  0.138
8  2015  0.0139 0.925
```

```
#Polity2
modelset %>% filter(YEAR < 2020) %>% group_by(YEAR) %>%
  summarize(cor = cor(N_EMBASSY,POLITY2, use = "pairwise.complete.obs") , pval = cor.test(N_E
```

```
# A tibble: 8 x 3
  YEAR      cor    pval
<dbl> <dbl> <dbl>
1  1965 -0.222  0.186
2  1975 -0.305  0.0370
3  1985 -0.199  0.170
4  1995 -0.284  0.0437
5  2000 -0.253  0.0735
6  2005 -0.209  0.141
7  2010 -0.198  0.164
8  2015 -0.0826 0.560
```

```
modelset %>% filter(YEAR < 2020) %>% group_by(YEAR) %>%
  summarize(cor = cor(N_EMBASSY,POLITY2_lag1, use = "pairwise.complete.obs") , pval = cor.test
```

```
# A tibble: 8 x 3
  YEAR      cor    pval
<dbl> <dbl> <dbl>
1  1965 -0.174  0.310
2  1975 -0.273  0.0762
3  1985 -0.220  0.129
```

```

4 1995 -0.323 0.0207
5 2000 -0.251 0.0782
6 2005 -0.222 0.117
7 2010 -0.180 0.206
8 2015 -0.110 0.438

```

```

modelset %>% filter(YEAR < 2020) %>% group_by(YEAR) %>%
  summarize(cor = cor(N_EMBASSY,POLITY2_lag2, use = "pairwise.complete.obs") , pval = cor.test

```

```

# A tibble: 8 x 3
  YEAR      cor    pval
<dbl>   <dbl> <dbl>
1 1965 -0.202 0.252
2 1975 -0.302 0.0523
3 1985 -0.0818 0.576
4 1995 -0.303 0.0308
5 2000 -0.217 0.127
6 2005 -0.246 0.0825
7 2010 -0.206 0.148
8 2015 -0.104 0.467

```

```

modelset %>% filter(YEAR < 2020) %>% group_by(YEAR) %>%
  summarize(cor = cor(N_EMBASSY,POLITY2_lag3, use = "pairwise.complete.obs") , pval = cor.test

```

```

# A tibble: 8 x 3
  YEAR      cor    pval
<dbl>   <dbl> <dbl>
1 1965 -0.238 0.183
2 1975 -0.350 0.0232
3 1985 -0.0842 0.565
4 1995 -0.240 0.0933
5 2000 -0.278 0.0486
6 2005 -0.265 0.0601
7 2010 -0.213 0.134
8 2015 -0.0112 0.938

```

```

#v2x_Libdem
modelset %>% group_by(YEAR) %>%
  summarize(cor = cor(N_EMBASSY,v2x_LIBDEM, use = "pairwise.complete.obs") , pval = cor.test

```

```
# A tibble: 10 x 3
  YEAR      cor    pval
  <dbl>   <dbl> <dbl>
1  1965 -0.117  0.496
2  1975 -0.143  0.331
3  1985 -0.157  0.270
4  1995 -0.277  0.0445
5  2000 -0.133  0.341
6  2005 -0.162  0.248
7  2010 -0.133  0.344
8  2015 -0.0577 0.679
9  2020 -0.113  0.416
10 2024 -0.137  0.323
```

```
modelset %>% group_by(YEAR) %>%
  summarize(cor = cor(N_EMBASSY,v2x_LIBDEM_lag1, use = "pairwise.complete.obs") , pval = cor
```

```
# A tibble: 10 x 3
  YEAR      cor    pval
  <dbl>   <dbl> <dbl>
1  1965 -0.0795 0.650
2  1975 -0.182  0.242
3  1985 -0.158  0.268
4  1995 -0.326  0.0173
5  2000 -0.158  0.260
6  2005 -0.151  0.282
7  2010 -0.147  0.294
8  2015 -0.0686 0.622
9  2020 -0.103  0.459
10 2024 -0.169  0.223
```

```
modelset %>% group_by(YEAR) %>%
  summarize(cor = cor(N_EMBASSY,v2x_LIBDEM_lag2, use = "pairwise.complete.obs") , pval = cor
```

```
# A tibble: 10 x 3
  YEAR      cor    pval
  <dbl>   <dbl> <dbl>
1  1965 -0.151  0.395
2  1975 -0.203  0.198
3  1985 -0.106  0.458
4  1995 -0.331  0.0155
```

5	2000	-0.214	0.123
6	2005	-0.131	0.350
7	2010	-0.165	0.237
8	2015	-0.0189	0.892
9	2020	-0.104	0.455
10	2024	-0.178	0.198

```
modelset %>% group_by(YEAR) %>%
  summarize(cor = cor(N_EMBASSY,v2x_LIBDEM_lag3, use = "pairwise.complete.obs") , pval = cor
```

```
# A tibble: 10 x 3
  YEAR      cor    pval
  <dbl>   <dbl> <dbl>
1  1965 -0.187  0.296
2  1975 -0.206  0.190
3  1985 -0.0854 0.551
4  1995 -0.333  0.0158
5  2000 -0.216  0.121
6  2005 -0.117  0.406
7  2010 -0.168  0.230
8  2015 -0.0160 0.908
9  2020 -0.101  0.468
10 2024 -0.169  0.222
```

```
modelset %>% filter(YEAR > 1965 & YEAR < 2024) %>% group_by(YEAR) %>%
  summarize(cor = cor(N_EMBASSY,TOTLIB1, use = "pairwise.complete.obs") , pval = cor.test(N_L
```

```
# A tibble: 8 x 3
  YEAR      cor    pval
  <dbl>   <dbl> <dbl>
1  1975  0.218  0.141
2  1985 -0.0334 0.816
3  1995  0.372  0.00612
4  2000  0.242  0.0812
5  2005  0.208  0.135
6  2010  0.217  0.118
7  2015  0.104  0.455
8  2020  0.180  0.194
```

```
modelset %>% filter(YEAR > 1965 & YEAR < 2024) %>% group_by(YEAR) %>%
  summarize(cor = cor(N_EMBASSY,TOTLIB1_lag1, use = "pairwise.complete.obs") , pval = cor.test
```

```
# A tibble: 8 x 3
  YEAR      cor    pval
<dbl>   <dbl>  <dbl>
1  1975  0.170  0.277
2  1985 -0.00192 0.989
3  1995  0.387  0.00423
4  2000  0.222  0.110
5  2005  0.257  0.0632
6  2010  0.234  0.0915
7  2015  0.106  0.446
8  2020  0.175  0.207
```

```
modelset %>% filter(YEAR > 1965 & YEAR < 2024) %>% group_by(YEAR) %>%
  summarize(cor = cor(N_EMBASSY,TOTLIB1_lag2, use = "pairwise.complete.obs") , pval = cor.test
```

```
# A tibble: 8 x 3
  YEAR      cor    pval
<dbl>   <dbl>  <dbl>
1  1975  0.329  0.0336
2  1985 -0.0483 0.737
3  1995  0.426  0.00148
4  2000  0.290  0.0349
5  2005  0.233  0.0931
6  2010  0.254  0.0665
7  2015  0.0823 0.554
8  2020  0.163  0.238
```

```
modelset %>% filter(YEAR > 1965 & YEAR < 2024) %>% group_by(YEAR) %>%
  summarize(cor = cor(N_EMBASSY,TOTLIB1_lag3, use = "pairwise.complete.obs") , pval = cor.test
```

```
# A tibble: 8 x 3
  YEAR      cor    pval
<dbl>   <dbl>  <dbl>
1  1975  0.291  0.0613
2  1985 -0.0216 0.880
3  1995  0.334  0.0155
4  2000  0.303  0.0277
```

Table 1: Reduced Model Parameters for **Democracy Scores** and Their **Lags**

		$\hat{\beta}$	$e^{\hat{\beta}}$	p	$\hat{\beta} : 95\%CI$	$e^{\hat{\beta}} : 95\%CI$
lag0	TOTLIB1	-0.006	0.994	0.304	-0.016, 0.005	0.984, 1.005
	POLITY	-0.015	0.985	0.000	-0.021, -0.009	0.979, 0.991
	POLITY2	-0.014	0.986	0.000	-0.021, -0.008	0.979, 0.992
	LIBDEM	0.114	1.121	0.245	-0.078, 0.307	0.925, 1.359
lag1	TOTLIB1	-0.004	0.996	0.489	-0.014, 0.007	0.986, 1.007
	POLITY	-0.015	0.985	0.000	-0.021, -0.008	0.979, 0.992
	POLITY2	-0.013	0.987	0.000	-0.019, -0.007	0.981, 0.993
	LIBDEM	0.105	1.111	0.292	-0.09, 0.3	0.914, 1.35
lag2	TOTLIB1	0.001	1.001	0.892	-0.01, 0.011	0.99, 1.011
	POLITY	-0.013	0.987	0.000	-0.019, -0.008	0.981, 0.992
	POLITY2	-0.012	0.988	0.000	-0.018, -0.007	0.982, 0.993
	LIBDEM	0.103	1.108	0.282	-0.085, 0.291	0.919, 1.338
lag3	TOTLIB1	0.000	1.000	0.980	-0.01, 0.01	0.99, 1.01
	POLITY	-0.012	0.988	0.000	-0.018, -0.006	0.982, 0.994
	POLITY2	-0.011	0.989	0.000	-0.017, -0.006	0.983, 0.994
	LIBDEM	0.107	1.113	0.265	-0.081, 0.295	0.922, 1.343

5 2005 0.270 0.0510
6 2010 0.296 0.0317
7 2015 0.0873 0.530
8 2020 0.178 0.199

Appendix

```
#Embassy models
summary(mod_TOTLIB1)
```

```
Generalized linear mixed model fit by maximum likelihood (Laplace
Approximation) [glmerMod]
Family: poisson ( log )
Formula: N_EMBASSY ~ TOTLIB1 + (1 | COUNTRY) + (1 | YEAR)
Data: modelset
Control: glmerControl(optimizer = "bobyqa")
```

Table 2: Full Model Parameters for **Democracy Scores** and Their **Lags**

		$\hat{\beta}$	$e^{\hat{\beta}}$	p	$\hat{\beta} : 95\%CI$	$e^{\hat{\beta}} : 95\%CI$
lag0	TOTLIB1	-0.002	0.998	0.678	-0.013, 0.009	0.987, 1.009
	POLITY	-0.010	0.990	0.003	-0.017, -0.004	0.983, 0.996
	POLITY2	-0.010	0.990	0.004	-0.016, -0.003	0.984, 0.997
	LIBDEM	0.106	1.112	0.311	-0.098, 0.311	0.907, 1.365
lag1	TOTLIB1	-0.004	0.996	0.526	-0.015, 0.008	0.985, 1.008
	POLITY	-0.010	0.990	0.002	-0.017, -0.004	0.983, 0.996
	POLITY2	-0.009	0.991	0.010	-0.015, -0.002	0.985, 0.998
	LIBDEM	0.120	1.127	0.254	-0.086, 0.326	0.918, 1.385
lag2	TOTLIB1	0.000	1.000	0.992	-0.01, 0.01	0.99, 1.01
	POLITY	-0.010	0.990	0.001	-0.016, -0.004	0.984, 0.996
	POLITY2	-0.009	0.991	0.003	-0.015, -0.003	0.985, 0.997
	LIBDEM	0.119	1.126	0.239	-0.079, 0.317	0.924, 1.373
lag3	TOTLIB1	-0.001	0.999	0.819	-0.012, 0.009	0.988, 1.009
	POLITY	-0.010	0.990	0.001	-0.016, -0.004	0.984, 0.996
	POLITY2	-0.009	0.991	0.002	-0.015, -0.003	0.985, 0.997
	LIBDEM	0.111	1.117	0.272	-0.086, 0.309	0.918, 1.362

Table 3: Delta Response Reduced Model for **Democracy Scores** and Their **Lags**

		$\hat{\beta}$	p	$\hat{\beta} : 95\%CI$
lag0	TOTLIB1	-0.060	0.614	-0.29, 0.175
	POLITY	-0.151	0.046	-0.303, -0.004
	POLITY2	-0.152	0.030	-0.294, -0.016
	LIBDEM	1.217	0.605	-3.499, 5.792
lag1	TOTLIB1	0.169	0.172	-0.072, 0.411
	POLITY	0.036	0.618	-0.105, 0.18
	POLITY2	0.043	0.516	-0.087, 0.175
	LIBDEM	-0.623	0.818	-5.932, 4.685
lag2	TOTLIB1	-0.167	0.200	-0.507, 0.151
	POLITY	0.060	0.430	-0.088, 0.207
	POLITY2	-0.012	0.000	-0.018, -0.007
	LIBDEM	2.043	0.502	-3.89, 8.018
lag3	TOTLIB1	-0.013	0.930	-0.304, 0.283
	POLITY	0.067	0.426	-0.102, 0.231
	POLITY2	0.042	0.586	-0.112, 0.192
	LIBDEM	3.859	0.313	-3.847, 11.257

Table 4: Delta Response Full Model for **Democracy Scores** and Their **Lags**

		$\hat{\beta}$	p	$\hat{\beta} : 95\%CI$
lag0	TOTLIB1	-0.080	0.500	-0.307, 0.158
	POLITY	-0.122	0.111	-0.274, 0.022
	POLITY2	-0.115	0.105	-0.255, 0.019
	LIBDEM	4.455	0.098	-0.869, 9.559
lag1	TOTLIB1	0.084	0.488	-0.147, 0.327
	POLITY	0.032	0.653	-0.105, 0.175
	POLITY2	0.048	0.465	-0.08, 0.177
	LIBDEM	0.494	0.864	-4.999, 6.157
lag2	TOTLIB1	-0.231	0.102	-0.599, -0.012
	POLITY	0.002	0.982	-0.152, 0.145
	POLITY2	-0.020	0.777	-0.16, 0.114
	LIBDEM	2.968	0.393	-3.695, 9.805
lag3	TOTLIB1	0.146	0.350	-0.152, 0.445
	POLITY	-0.007	0.930	-0.175, 0.149
	POLITY2	-0.011	0.889	-0.159, 0.13
	LIBDEM	2.265	0.575	-5.75, 9.821

Table 5: Delta Response Delta Predictors Full Model for **Democracy Scores** and Their **Lags**

		$\hat{\beta}$	p	$\hat{\beta} : 95\%CI$
lag0	TOTLIB1	-0.079	0.518	-0.309, 0.166
	POLITY	-0.095	0.212	-0.248, 0.049
	POLITY2	-0.099	0.165	-0.244, 0.035
	LIBDEM	3.722	0.171	-1.705, 8.919
lag1	TOTLIB1	0.081	0.518	-0.16, 0.326
	POLITY	0.042	0.573	-0.1, 0.189
	POLITY2	0.040	0.562	-0.092, 0.174
	LIBDEM	0.347	0.906	-5.276, 6.163
lag2	TOTLIB1	-0.263	0.082	-0.574, 0.018
	POLITY	0.031	0.694	-0.122, 0.182
	POLITY2	-0.003	0.970	-0.144, 0.137
	LIBDEM	3.153	0.389	-3.808, 10.416
lag3	TOTLIB1	0.090	0.579	-0.222, 0.402
	POLITY	0.028	0.748	-0.144, 0.195
	POLITY2	0.011	0.893	-0.145, 0.162
	LIBDEM	3.908	0.349	-4.15, 12.22

AIC	BIC	logLik	-2*log(L)	df.resid
2775.4	2791.6	-1383.7	2767.4	414

Scaled residuals:

Min	1Q	Median	3Q	Max
-4.2204	-0.5436	0.0196	0.4853	3.6965

Random effects:

Groups	Name	Variance	Std.Dev.
COUNTRY	(Intercept)	0.50528	0.7108
YEAR	(Intercept)	0.02908	0.1705

Number of obs: 418, groups: COUNTRY, 54; YEAR, 8

Fixed effects:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	3.164714	0.125803	25.156	<2e-16 ***
TOTLIB1	-0.005548	0.005398	-1.028	0.304

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

(Intr)
TOTLIB1 -0.412

```
summary(mod1_TOTLIB1)
```

Generalized linear mixed model fit by maximum likelihood (Laplace Approximation) [glmerMod]
Family: poisson (log)
Formula: N_EMBASSY ~ TOTLIB1 + logGNI + logPOPULATN + REGION + COL2 + factor(IDEOLOGY) + factor(AFR_HIST_PERIOD) + (1 | COUNTRY) + (1 | YEAR)
Data: modelset
Control: glmerControl(optimizer = "bobyqa")

AIC	BIC	logLik	-2*log(L)	df.resid
2418.0	2473.3	-1195.0	2390.0	370

Scaled residuals:

Min	1Q	Median	3Q	Max
-4.3722	-0.5446	-0.0311	0.4472	3.2687

Groups	Name	Variance	Std.Dev.
COUNTRY	(Intercept)	0.0708070	0.26610
YEAR	(Intercept)	0.0006951	0.02637

Number of obs: 384, groups: COUNTRY, 54; YEAR, 8

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	1.073997	0.268993	3.993	6.53e-05	***
TOTLIB1	-0.002307	0.005555	-0.415	0.6779	
logGNI	0.282964	0.048238	5.866	4.46e-09	***
logPOPULATN	0.468052	0.075011	6.240	4.38e-10	***
REGIONCentral	-0.628896	0.156736	-4.012	6.01e-05	***
REGIONEast	-0.950200	0.153107	-6.206	5.43e-10	***
REGIONSouth	-0.860868	0.153885	-5.594	2.22e-08	***
REGIONWest	-0.655524	0.146134	-4.486	7.27e-06	***
COL2Not Colony	0.042069	0.293838	0.143	0.8862	
factor(IDEOLOGY)2	0.020061	0.062479	0.321	0.7481	
factor(IDEOLOGY)3	-0.155148	0.090208	-1.720	0.0855	.
factor(AFR_HIST_PERIOD)Post-Cold War	-0.054966	0.043409	-1.266	0.2054	

Correlation of Fixed Effects:

29

```

REGIONSouth
REGIONWest
COL2NotClny
f(IDEOLOGY)2
f(IDEOLOGY)3 0.125
f(AFR_HISTW 0.197      0.177

```

```
summary(mod_POLITY)
```

```

Generalized linear mixed model fit by maximum likelihood (Laplace
Approximation) [glmerMod]
Family: poisson ( log )
Formula: N_EMBASSY ~ POLITY + (1 | COUNTRY) + (1 | YEAR)
Data: modelset
Control: glmerControl(optimizer = "bobyqa")

```

AIC	BIC	logLik	-2*log(L)	df.resid
2536.2	2551.8	-1264.1	2528.2	363

Scaled residuals:

Min	1Q	Median	3Q	Max
-4.9175	-0.5623	-0.0293	0.5593	4.0286

Random effects:

Groups	Name	Variance	Std.Dev.
COUNTRY	(Intercept)	0.4320	0.6573
YEAR	(Intercept)	0.1108	0.3329

Number of obs: 367, groups: COUNTRY, 51; YEAR, 8

Fixed effects:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	3.023268	0.150032	20.151	< 2e-16 ***
POLITY	-0.015025	0.003256	-4.614	3.94e-06 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

```

(Intr)
POLITY 0.030

```

```
summary(mod1_POLITY)
```

Generalized linear mixed model fit by maximum likelihood (Laplace

Approximation) [glmerMod]

Family: poisson (log)

Formula:

N_EMBASSY ~ POLITY + logGNI + logPOPULATN + REGION + COL2 + factor(IDEOLOGY) +
factor(AFR_HIST_PERIOD) + (1 | COUNTRY) + (1 | YEAR)

Data: modelset

Control: glmerControl(optimizer = "bobyqa")

AIC	BIC	logLik	-2*log(L)	df.resid
2123.0	2176.3	-1047.5	2095.0	319

Scaled residuals:

Min	1Q	Median	3Q	Max
-3.9380	-0.5663	-0.0497	0.4440	3.7553

Random effects:

Groups	Name	Variance	Std.Dev.
--------	------	----------	----------

COUNTRY	(Intercept)	0.058259	0.24137
---------	-------------	----------	---------

YEAR	(Intercept)	0.004488	0.06699
------	-------------	----------	---------

Number of obs: 333, groups: COUNTRY, 51; YEAR, 8

Fixed effects:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	0.62085	0.29205	2.126	0.03352 *
POLITY	-0.01023	0.00340	-3.008	0.00263 **
logGNI	0.41881	0.05719	7.323	2.43e-13 ***
logPOPULATN	0.38323	0.07987	4.798	1.60e-06 ***
REGIONCentral	-0.46918	0.14983	-3.131	0.00174 **
REGIONEast	-0.76173	0.14938	-5.099	3.41e-07 ***
REGIONSouth	-0.72259	0.14514	-4.979	6.40e-07 ***
REGIONWest	-0.43827	0.14028	-3.124	0.00178 **
COL2Not Colony	0.03424	0.27267	0.126	0.90008
factor(IDEOLOGY)2	0.09713	0.06324	1.536	0.12454
factor(IDEOLOGY)3	-0.16843	0.09066	-1.858	0.06320 .
factor(AFR_HIST_PERIOD)Post-Cold War	0.03321	0.06973	0.476	0.63383

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

```

(Intr) POLITY logGNI lPOPUL REGIONC REGIONE REGIONS REGIONW COL2NC
POLITY          0.090
logGNI          -0.129  0.077
logPOPULATN    -0.645 -0.056 -0.592
REGIONCntrl    -0.545 -0.002  0.244  0.056
REGIONEast     -0.480 -0.076  0.262 -0.020  0.663
REGIONSouth    -0.517 -0.124  0.166  0.071  0.667  0.664
REGIONWest     -0.563 -0.093  0.309  0.002  0.726  0.723  0.724
COL2NotClny    0.188  0.029  0.053 -0.204 -0.041 -0.208 -0.039 -0.035
f(IDEOLOGY)2   -0.112  0.226  0.062 -0.010  0.132  0.104  0.082  0.106 -0.003
f(IDEOLOGY)3   -0.033  0.082 -0.066  0.062 -0.008 -0.028 -0.024 -0.022 -0.040
f(AFR_HISTW    0.165 -0.277 -0.428  0.027 -0.144 -0.121 -0.083 -0.146  0.027
f(IDEOLOGY)2 f(IDEOLOGY)3
POLITY
logGNI
logPOPULATN
REGIONCntrl
REGIONEast
REGIONSouth
REGIONWest
COL2NotClny
f(IDEOLOGY)2
f(IDEOLOGY)3  0.118
f(AFR_HISTW  0.104      0.099

```

```
summary(mod_POLITY2)
```

```

Generalized linear mixed model fit by maximum likelihood (Laplace
Approximation) [glmerMod]
Family: poisson ( log )
Formula: N_EMBASSY ~ POLITY2 + (1 | COUNTRY) + (1 | YEAR)
Data: modelset
Control: glmerControl(optimizer = "bobyqa")

```

AIC	BIC	logLik	-2*log(L)	df.resid
2673.2	2689.1	-1332.6	2665.2	385

Scaled residuals:

Min	1Q	Median	3Q	Max
-4.9490	-0.5773	-0.0587	0.5352	4.3477

Random effects:

Groups	Name	Variance	Std.Dev.
COUNTRY	(Intercept)	0.4364	0.6606
YEAR	(Intercept)	0.1073	0.3275

Number of obs: 389, groups: COUNTRY, 52; YEAR, 8

Fixed effects:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	3.00369	0.14833	20.250	< 2e-16 ***
POLITY2	-0.01431	0.00316	-4.528	5.97e-06 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

(Intr)
POLITY2 0.030

```
summary(mod1_POLITY2)
```

Generalized linear mixed model fit by maximum likelihood (Laplace
Approximation) [glmerMod]
Family: poisson (log)
Formula: N_EMBASSY ~ POLITY2 + logGNI + logPOPULATN + REGION + COL2 +
factor(IDEOLOGY) + factor(AFR_HIST_PERIOD) + (1 | COUNTRY) +
(1 | YEAR)
Data: modelset
Control: glmerControl(optimizer = "bobyqa")

AIC	BIC	logLik	-2*log(L)	df.resid
2228.4	2282.4	-1100.2	2200.4	336

Scaled residuals:

Min	1Q	Median	3Q	Max
-4.0050	-0.5398	-0.0500	0.4520	3.6982

Random effects:

Groups	Name	Variance	Std.Dev.
COUNTRY	(Intercept)	0.063474	0.2519
YEAR	(Intercept)	0.004422	0.0665

Number of obs: 350, groups: COUNTRY, 52; YEAR, 8

Fixed effects:

Estimate	Std. Error	z value	Pr(> z)
----------	------------	---------	----------

(Intercept)	0.709424	0.297914	2.381	0.017251	*
POLITY2	-0.009667	0.003318	-2.914	0.003570	**
logGNI	0.388138	0.054663	7.101	1.24e-12	***
logPOPULATN	0.403862	0.080714	5.004	5.63e-07	***
REGIONCentral	-0.533005	0.153510	-3.472	0.000516	***
REGIONEast	-0.852376	0.151273	-5.635	1.75e-08	***
REGIONSouth	-0.772893	0.149214	-5.180	2.22e-07	***
REGIONWest	-0.501687	0.143128	-3.505	0.000456	***
COL2Not Colony	0.065509	0.282574	0.232	0.816672	
factor(IDEOLOGY)2	0.040643	0.058763	0.692	0.489160	
factor(IDEOLOGY)3	-0.177004	0.090315	-1.960	0.050014	.
factor(AFR_HIST_PERIOD)Post-Cold War	0.034495	0.068483	0.504	0.614469	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

	(Intr)	POLITY	logGNI	lPOPUL	REGIONC	REGIONE	REGIONS	REGIONW	COL2NC
POLITY2	0.089								
logGNI	-0.101	0.081							
logPOPULATN	-0.681	-0.061	-0.570						
REGIONCntrl	-0.535	0.000	0.216	0.076					
REGIONEast	-0.478	-0.071	0.233	0.005	0.665				
REGIONSouth	-0.510	-0.116	0.140	0.091	0.662	0.666			
REGIONWest	-0.553	-0.086	0.274	0.030	0.720	0.724	0.719		
COL2NotClny	0.191	0.028	0.050	-0.205	-0.042	-0.202	-0.040	-0.037	
f(IDEOLOGY)2	-0.054	0.258	0.000	0.002	0.077	0.052	0.034	0.046	0.001
f(IDEOLOGY)3	-0.028	0.093	-0.063	0.057	-0.011	-0.028	-0.026	-0.024	-0.039
f(AFR_HISTW	0.185	-0.266	-0.418	-0.011	-0.145	-0.120	-0.086	-0.145	0.035
	f(IDEOLOGY)2	f(IDEOLOGY)3							

POLITY2		
logGNI		
logPOPULATN		
REGIONCntrl		
REGIONEast		
REGIONSouth		
REGIONWest		
COL2NotClny		
f(IDEOLOGY)2		
f(IDEOLOGY)3	0.110	
f(AFR_HISTW	0.108	0.086

```
summary(mod_v2x_LIBDEM)
```

```
Generalized linear mixed model fit by maximum likelihood (Laplace
Approximation) [glmerMod]
Family: poisson ( log )
Formula: N_EMBASSY ~ v2x_LIBDEM + (1 | COUNTRY) + (1 | YEAR)
Data: modelset
Control: glmerControl(optimizer = "bobyqa")
```

AIC	BIC	logLik	-2*log(L)	df.resid
3472.2	3489.1	-1732.1	3464.2	505

Scaled residuals:

Min	1Q	Median	3Q	Max
-5.1678	-0.5799	-0.0057	0.4953	5.1574

Random effects:

Groups	Name	Variance	Std.Dev.
COUNTRY	(Intercept)	0.47283	0.6876
YEAR	(Intercept)	0.09601	0.3098

Number of obs: 509, groups: COUNTRY, 54; YEAR, 10

Fixed effects:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	3.05262	0.13774	22.162	<2e-16 ***
v2x_LIBDEM	0.11410	0.09811	1.163	0.245

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

	(Intr)
v2x_LIBDEM	-0.161

```
summary(mod1_v2x_LIBDEM)
```

```
Generalized linear mixed model fit by maximum likelihood (Laplace
Approximation) [glmerMod]
Family: poisson ( log )
Formula: N_EMBASSY ~ v2x_LIBDEM + logGNI + logPOPULATN + REGION + COL2 +
  factor(IDEOLOGY) + factor(AFR_HIST_PERIOD) + (1 | COUNTRY) +
  (1 | YEAR)
```

Data: modelset
Control: glmerControl(optimizer = "bobyqa")

AIC	BIC	logLik	-2*log(L)	df.resid
2603.4	2659.6	-1287.7	2575.4	395

Scaled residuals:

Min	1Q	Median	3Q	Max
-4.1489	-0.5532	-0.0172	0.4329	3.6618

Random effects:

Groups	Name	Variance	Std.Dev.
COUNTRY	(Intercept)	0.066823	0.25850
YEAR	(Intercept)	0.005327	0.07299

Number of obs: 409, groups: COUNTRY, 54; YEAR, 9

Fixed effects:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	0.79039	0.26824	2.947	0.003213 **
v2x_LIBDEM	0.10554	0.10419	1.013	0.311091
logGNI	0.34744	0.05100	6.812	9.62e-12 ***
logPOPULATN	0.44077	0.07165	6.152	7.66e-10 ***
REGIONCentral	-0.57381	0.15419	-3.721	0.000198 ***
REGIONEast	-0.89584	0.15044	-5.955	2.60e-09 ***
REGIONSouth	-0.83308	0.15051	-5.535	3.11e-08 ***
REGIONWest	-0.59178	0.14376	-4.116	3.85e-05 ***
COL2Not Colony	0.03781	0.28612	0.132	0.894881
factor(IDEOLOGY)2	0.08304	0.05383	1.543	0.122916
factor(IDEOLOGY)3	-0.13786	0.08876	-1.553	0.120399
factor(AFR_HIST_PERIOD)Post-Cold War	-0.01059	0.06936	-0.153	0.878655

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

	(Intr)	v2_LIB	logGNI	lPOPUL	REGIONC	REGIONE	REGIONS	REGIONW	COL2NC
v2x_LIBDEM	-0.079								
logGNI	-0.118	-0.086							
logPOPULATN	-0.638	0.105	-0.584						
REGIONCntrl	-0.577	0.011	0.209	0.079					
REGIONEast	-0.545	-0.030	0.207	0.045	0.689				
REGIONSouth	-0.504	-0.108	0.149	0.050	0.667	0.678			
REGIONWest	-0.547	-0.106	0.264	-0.006	0.722	0.733	0.718		
COL2NotClny	0.186	0.022	0.035	-0.198	-0.046	-0.198	-0.036	-0.035	

```

f(IDEOLOGY)2 -0.058  0.106 -0.060  0.027  0.045  0.039  0.026  0.026  0.001
f(IDEOLOGY)3 -0.038  0.054 -0.068  0.062 -0.011 -0.017 -0.020 -0.021 -0.033
f(AFR_HISTW  0.185 -0.158 -0.402 -0.014 -0.153 -0.137 -0.090 -0.137  0.042
      f(IDEOLOGY)2 f(IDEOLOGY)3
v2x_LIBDEM
logGNI
logPOPULATN
REGIONCntrl
REGIONEast
REGIONSouth
REGIONWest
COL2NotClny
f(IDEOLOGY)2
f(IDEOLOGY)3  0.091
f(AFR_HISTW  0.175      0.097

```

```

#Change in Embassy Models
summary(mod0_delta_TOTLIB1_PER_YEAR)

```

```

Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]
Formula: DELTA_N_EMBASSY_PER_YEAR ~ DELTA_TOTLIB1_PER_YEAR + (1 | COUNTRY) +
      (1 | YEAR)
Data: modelset_delta

```

REML criterion at convergence: 761.8

```

Scaled residuals:
      Min       1Q   Median       3Q      Max
-5.0764 -0.5325 -0.1444  0.3643  4.2541

```

```

Random effects:
Groups   Name              Variance Std.Dev.
COUNTRY  (Intercept)  0.04627  0.2151
YEAR     (Intercept)  0.05134  0.2266
Residual                    0.41749  0.6461
Number of obs: 364, groups:  COUNTRY, 54; YEAR, 7

```

```

Fixed effects:
              Estimate Std. Error      df t value Pr(>|t|)
(Intercept)    0.41635    0.09679   7.26138  4.302  0.00327 **
DELTA_TOTLIB1_PER_YEAR -0.05969    0.11826 326.54277 -0.505  0.61409

```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

(Intr)

DELTA_TOTLI 0.039

```
summary(mod1_delta_TOTLIB1_PER_YEAR)
```

Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]

Formula: DELTA_N_EMBASSY_PER_YEAR ~ DELTA_TOTLIB1_PER_YEAR + logGNI +
logPOPULATN + REGION + COL2 + factor(IDEOLOGY) + factor(AFR_HIST_PERIOD) +
(1 | COUNTRY) + (1 | YEAR)

Data: modelset_delta

REML criterion at convergence: 686.3

Scaled residuals:

Min	1Q	Median	3Q	Max
-5.4579	-0.5419	-0.0860	0.4279	4.0387

Random effects:

Groups	Name	Variance	Std.Dev.
COUNTRY	(Intercept)	0.0149	0.1221
YEAR	(Intercept)	0.0464	0.2154
Residual		0.3918	0.6259

Number of obs: 341, groups: COUNTRY, 53; YEAR, 7

Fixed effects:

	Estimate	Std. Error	df	t value
(Intercept)	-0.65706	0.39632	29.90572	-1.658
DELTA_TOTLIB1_PER_YEAR	-0.08028	0.11899	300.20222	-0.675
logGNI	0.38717	0.10589	68.58931	3.656
logPOPULATN	-0.10862	0.09889	57.82290	-1.098
REGIONCentral	0.04897	0.17628	44.29902	0.278
REGIONEast	0.19299	0.17205	44.76740	1.122
REGIONSouth	0.20570	0.16352	43.07112	1.258
REGIONWest	0.14595	0.16815	45.49572	0.868
COL2Not Colony	-0.16080	0.29182	38.25036	-0.551
factor(IDEOLOGY)2	-0.07092	0.26038	321.38626	-0.272
factor(IDEOLOGY)3	0.13749	0.28617	321.78245	0.480

```

factor(AFR_HIST_PERIOD)Post-Cold War -0.12110    0.26443    6.06860   -0.458
                                         Pr(>|t|)
(Intercept)                           0.107790
DELTA_TOTLIB1_PER_YEAR                 0.500381
logGNI                                0.000497 ***
logPOPULATN                           0.276581
REGIONCentral                         0.782464
REGIONEast                           0.267952
REGIONSouth                          0.215173
REGIONWest                           0.389963
COL2Not Colony                       0.584816
factor(IDEOLOGY)2                     0.785500
factor(IDEOLOGY)3                     0.631225
factor(AFR_HIST_PERIOD)Post-Cold War 0.662906
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

Correlation of Fixed Effects:

```

              (Intr) DELTA_ logGNI lPOPUL REGIONC REGIONE REGIONS REGIONW COL2NC
DELTA_TOTLI   0.020
logGNI        -0.316 -0.057
logPOPULATN  -0.150  0.045 -0.790
REGIONCntrl  -0.535 -0.082  0.431 -0.190
REGIONEast   -0.498 -0.050  0.377 -0.165  0.743
REGIONSouth  -0.467 -0.033  0.303 -0.121  0.728  0.719
REGIONWest   -0.528 -0.006  0.498 -0.275  0.794  0.775  0.759
COL2NotClny   0.137 -0.016  0.072 -0.199 -0.024 -0.181 -0.020 -0.007
f(IDEOLOGY)2 -0.157 -0.032 -0.038  0.055  0.140  0.135  0.119  0.097 -0.022
f(IDEOLOGY)3 -0.125 -0.048 -0.016  0.032  0.001 -0.013 -0.017 -0.016 -0.125
f(AFR_HISTW   -0.496  0.036 -0.158  0.104 -0.060 -0.061 -0.052 -0.072 -0.017
              f(IDEOLOGY)2 f(IDEOLOGY)3
DELTA_TOTLI
logGNI
logPOPULATN
REGIONCntrl
REGIONEast
REGIONSouth
REGIONWest
COL2NotClny
f(IDEOLOGY)2
f(IDEOLOGY)3  0.126
f(AFR_HISTW   0.137      0.172

```

```
summary(mod0_delta_POLITY_PER_YEAR)
```

```
Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]
Formula: DELTA_N_EMBASSY_PER_YEAR ~ DELTA_POLITY_PER_YEAR + (1 | COUNTRY) +
(1 | YEAR)
Data: modelset_delta
```

REML criterion at convergence: 687.8

Scaled residuals:

Min	1Q	Median	3Q	Max
-5.0134	-0.5004	-0.1430	0.3870	3.7323

Random effects:

Groups	Name	Variance	Std.Dev.
COUNTRY	(Intercept)	0.07530	0.2744
YEAR	(Intercept)	0.09609	0.3100
Residual		0.47319	0.6879

Number of obs: 304, groups: COUNTRY, 51; YEAR, 7

Fixed effects:

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	0.53999	0.12997	6.99608	4.155	0.00427 **
DELTA_POLITY_PER_YEAR	-0.15065	0.07529	274.17066	-2.001	0.04640 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

	(Intr)
DELTA_POLIT	-0.065

```
summary(mod1_delta_POLITY_PER_YEAR)
```

```
Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]
Formula:
DELTA_N_EMBASSY_PER_YEAR ~ DELTA_POLITY_PER_YEAR + logGNI + logPOPULATN +
REGION + COL2 + factor(IDEOLOGY) + factor(AFR_HIST_PERIOD) +
(1 | COUNTRY) + (1 | YEAR)
Data: modelset_delta
```


REML criterion at convergence: 622.3

Scaled residuals:

Min	1Q	Median	3Q	Max
-5.2767	-0.5509	-0.1040	0.4192	3.5175

Random effects:

Groups	Name	Variance	Std.Dev.
COUNTRY	(Intercept)	0.01092	0.1045
YEAR	(Intercept)	0.08453	0.2907
Residual		0.45566	0.6750

Number of obs: 287, groups: COUNTRY, 50; YEAR, 7

Fixed effects:

	Estimate	Std. Error	df	t value
(Intercept)	-0.640147	0.444543	34.023115	-1.440
DELTA_POLITY_PER_YEAR	-0.121696	0.076166	255.878346	-1.598
logGNI	0.470514	0.124574	46.292164	3.777
logPOPULATN	-0.079876	0.125114	39.189522	-0.638
REGIONCentral	0.119526	0.193561	28.095601	0.618
REGIONEast	0.160639	0.194709	29.333823	0.825
REGIONSouth	0.231321	0.179089	27.508842	1.292
REGIONWest	0.125942	0.185194	28.576979	0.680
COL2Not Colony	-0.377311	0.325690	30.283694	-1.158
factor(IDEOLOGY)2	0.032697	0.248559	236.213908	0.132
factor(IDEOLOGY)3	0.008992	0.239896	251.213349	0.037
factor(AFR_HIST_PERIOD)Post-Cold War	-0.576184	0.270491	5.778399	-2.130

	Pr(> t)
(Intercept)	0.159003
DELTA_POLITY_PER_YEAR	0.111326
logGNI	0.000452 ***
logPOPULATN	0.526911
REGIONCentral	0.541870
REGIONEast	0.416018
REGIONSouth	0.207224
REGIONWest	0.501944
COL2Not Colony	0.255716
factor(IDEOLOGY)2	0.895455
factor(IDEOLOGY)3	0.970131
factor(AFR_HIST_PERIOD)Post-Cold War	0.078954 .

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

	(Intr)	DELTA_	logGNI	lPOPUL	REGIONC	REGIONE	REGIONS	REGIONW	COL2NC
DELTA_POLIT	0.033								
logGNI	-0.264	0.066							
logPOPULATN	-0.283	-0.086	-0.772						
REGIONCntrl	-0.555	0.066	0.380	-0.103					
REGIONEast	-0.451	0.067	0.390	-0.188	0.702				
REGIONSouth	-0.476	0.055	0.277	-0.066	0.711	0.684			
REGIONWest	-0.557	0.024	0.493	-0.218	0.783	0.752	0.748		
COL2NotClny	0.151	0.050	0.094	-0.213	-0.026	-0.186	-0.018	-0.007	
f(IDEOLOGY)2	-0.141	0.054	-0.045	0.064	0.178	0.154	0.130	0.133	-0.021
f(IDEOLOGY)3	-0.091	-0.025	-0.031	0.048	-0.018	-0.056	-0.047	-0.023	-0.099
f(AFR_HISTW	-0.343	-0.108	-0.188	0.119	-0.072	-0.092	-0.078	-0.091	-0.027
	f(IDEOLOGY)2	f(IDEOLOGY)3							

DELTA_POLIT

logGNI

logPOPULATN

REGIONCntrl

REGIONEast

REGIONSouth

REGIONWest

COL2NotClny

f(IDEOLOGY)2

f(IDEOLOGY)3 0.117

f(AFR_HISTW 0.094 0.141

`summary(mod0_delta_POLITY2_PER_YEAR)`

Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]

Formula: DELTA_N_EMBASSY_PER_YEAR ~ DELTA_POLITY2_PER_YEAR + (1 | COUNTRY) +
(1 | YEAR)

Data: modelset_delta

REML criterion at convergence: 757.1

Scaled residuals:

Min	1Q	Median	3Q	Max
-4.9539	-0.5111	-0.1137	0.4121	3.7781

Random effects:

Groups	Name	Variance	Std.Dev.
COUNTRY	(Intercept)	0.06738	0.2596
YEAR	(Intercept)	0.10333	0.3214
Residual		0.47163	0.6868

Number of obs: 337, groups: COUNTRY, 51; YEAR, 7

Fixed effects:

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	0.5148	0.1326	6.7849	3.882	0.00642 **
DELTA_POLITY2_PER_YEAR	-0.1525	0.0701	303.6658	-2.175	0.03038 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

	(Intr)
DELTA_POLIT	-0.067

`summary(mod1_delta_POLITY2_PER_YEAR)`

Linear mixed model fit by REML. t-tests use Satterthwaite's method [lmerModLmerTest]
 Formula: DELTA_N_EMBASSY_PER_YEAR ~ DELTA_POLITY2_PER_YEAR + logGNI + logPOPULATN + REGION + COL2 + factor(IDEOLOGY) + factor(AFR_HIST_PERIOD) + (1 | COUNTRY) + (1 | YEAR)
 Data: modelset_delta

REML criterion at convergence: 675.6

Scaled residuals:

Min	1Q	Median	3Q	Max
-5.2467	-0.6070	-0.1027	0.4294	3.6496

Random effects:

Groups	Name	Variance	Std.Dev.
COUNTRY	(Intercept)	8.652e-09	9.301e-05
YEAR	(Intercept)	9.132e-02	3.022e-01
Residual		4.549e-01	6.745e-01

Number of obs: 315, groups: COUNTRY, 51; YEAR, 7

Fixed effects:

	Estimate	Std. Error	df	t value
(Intercept)	-0.431348	0.410132	48.176923	-1.052

DELTA_POLITY2_PER_YEAR	-0.115448	0.070917	301.664237	-1.628
logGNI	0.461297	0.112050	302.051787	4.117
logPOPULATN	-0.113690	0.112669	302.166755	-1.009
REGIONCentral	-0.001338	0.173901	300.047323	-0.008
REGIONEast	0.089351	0.174698	300.040018	0.511
REGIONSouth	0.203394	0.162779	299.144934	1.250
REGIONWest	0.069633	0.167575	300.916947	0.416
COL2Not Colony	-0.326459	0.301494	297.940303	-1.083
factor(IDEOLOGY)2	0.002456	0.243148	298.812557	0.010
factor(IDEOLOGY)3	0.021097	0.234498	298.688994	0.090
factor(AFR_HIST_PERIOD)Post-Cold War	-0.558357	0.275988	5.578399	-2.023

Pr(>|t|)

(Intercept)	0.2982
DELTA_POLITY2_PER_YEAR	0.1046
logGNI	4.96e-05 ***
logPOPULATN	0.3138
REGIONCentral	0.9939
REGIONEast	0.6094
REGIONSouth	0.2125
REGIONWest	0.6780
COL2Not Colony	0.2798
factor(IDEOLOGY)2	0.9919
factor(IDEOLOGY)3	0.9284
factor(AFR_HIST_PERIOD)Post-Cold War	0.0931 .

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

	(Intr)	DELTA_	logGNI	lPOPUL	REGIONC	REGIONE	REGIONS	REGIONW	COL2NC
DELTA_POLIT	0.012								
logGNI	-0.251	0.088							
logPOPULATN	-0.260	-0.098	-0.781						
REGIONCntrl	-0.518	0.090	0.409	-0.158					
REGIONEast	-0.438	0.092	0.394	-0.203	0.724				
REGIONSouth	-0.452	0.085	0.282	-0.085	0.719	0.695			
REGIONWest	-0.528	0.047	0.497	-0.239	0.797	0.765	0.752		
COL2NotClny	0.141	0.051	0.092	-0.205	-0.013	-0.165	-0.011	-0.002	
f(IDEOLOGY)2	-0.138	0.060	-0.033	0.050	0.177	0.163	0.129	0.136	-0.019
f(IDEOLOGY)3	-0.093	-0.013	-0.023	0.044	-0.019	-0.046	-0.047	-0.023	-0.110
f(AFR_HISTW	-0.410	-0.098	-0.156	0.101	-0.065	-0.076	-0.071	-0.078	-0.027

f(IDEOLOGY)2 f(IDEOLOGY)3

DELTA_POLIT

logGNI

```
logPOPULATN
REGIONCntrl
REGIONEast
REGIONSouth
REGIONWest
COL2NotClny
f(IDEOLOGY)2
f(IDEOLOGY)3 0.110
f(AFR_HISTW 0.083 0.125
```

```
summary(mod0_delta_v2x_LIBDEM_PER_YEAR)
```

```
Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]
```

```
Formula:
```

```
DELTA_N_EMBASSY_PER_YEAR ~ DELTA_v2x_LIBDEM_PER_YEAR + (1 | COUNTRY) +
(1 | YEAR)
```

```
Data: modelset_delta
```

```
REML criterion at convergence: 1006.3
```

```
Scaled residuals:
```

```
      Min       1Q   Median       3Q      Max
-4.7968 -0.5541 -0.1112  0.3922  3.8657
```

```
Random effects:
```

```
Groups   Name             Variance Std.Dev.
COUNTRY  (Intercept) 0.03548  0.1884
YEAR     (Intercept) 0.09425  0.3070
Residual                    0.48556  0.6968
```

```
Number of obs: 455, groups: COUNTRY, 54; YEAR, 9
```

```
Fixed effects:
```

```
              Estimate Std. Error      df t value Pr(>|t|)
(Intercept)      0.5293     0.1106   8.5852   4.786 0.00113 **
DELTA_v2x_LIBDEM_PER_YEAR  1.2168     2.3506  427.1648   0.518 0.60497
```

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Correlation of Fixed Effects:
```

```
      (Intr)
DELTA_2_LIB -0.022
```

```
summary(mod1_delta_v2x_LIBDEM_PER_YEAR)
```

```
Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]
Formula: DELTA_N_EMBASSY_PER_YEAR ~ DELTA_v2x_LIBDEM_PER_YEAR + logGNI +
      logPOPULATN + REGION + COL2 + factor(IDEOLOGY) + factor(AFR_HIST_PERIOD) +
      (1 | COUNTRY) + (1 | YEAR)
Data: modelset_delta
```

REML criterion at convergence: 776.4

Scaled residuals:

Min	1Q	Median	3Q	Max
-5.2621	-0.5656	-0.0843	0.4133	3.7264

Random effects:

Groups	Name	Variance	Std.Dev.
COUNTRY	(Intercept)	0.00000	0.0000
YEAR	(Intercept)	0.08371	0.2893
Residual		0.43661	0.6608

Number of obs: 373, groups: COUNTRY, 53; YEAR, 8

Fixed effects:

	Estimate	Std. Error	df	t value
(Intercept)	-0.35017	0.36273	43.08742	-0.965
DELTA_v2x_LIBDEM_PER_YEAR	4.45475	2.68374	359.63939	1.660
logGNI	0.43240	0.09953	358.94538	4.344
logPOPULATN	-0.12336	0.09296	360.60473	-1.327
REGIONCentral	0.09715	0.15778	358.07439	0.616
REGIONEast	0.16761	0.15526	357.39191	1.080
REGIONSouth	0.19971	0.14692	356.63089	1.359
REGIONWest	0.09714	0.15055	358.87484	0.645
COL2Not Colony	-0.15177	0.27210	354.96756	-0.558
factor(IDEOLOGY)2	0.08532	0.23702	355.91812	0.360
factor(IDEOLOGY)3	0.01849	0.22949	355.84078	0.081
factor(AFR_HIST_PERIOD)Post-Cold War	-0.54627	0.25868	6.74808	-2.112

Pr(>|t|)

(Intercept)	0.3397
DELTA_v2x_LIBDEM_PER_YEAR	0.0978 .
logGNI	1.82e-05 ***
logPOPULATN	0.1853
REGIONCentral	0.5385

```

REGIONEast                0.2811
REGIONSouth               0.1749
REGIONWest                0.5192
COL2Not Colony            0.5774
factor(IDEOLOGY)2         0.7191
factor(IDEOLOGY)3         0.9358
factor(AFR_HIST_PERIOD)Post-Cold War 0.0741 .

```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

```

              (Intr) DELTA_ logGNI lPOPUL REGIONC REGIONE REGIONS REGIONW COL2NC
DELTA_2_LIB   -0.033
logGNI        -0.302  0.048
logPOPULATN   -0.164 -0.027 -0.795
REGIONCntrl   -0.524  0.091  0.424 -0.185
REGIONEast    -0.475  0.085  0.369 -0.165  0.737
REGIONSouth   -0.446  0.056  0.293 -0.117  0.720  0.707
REGIONWest    -0.524  0.025  0.496 -0.270  0.796  0.768  0.751
COL2NotClny    0.136 -0.016  0.073 -0.194 -0.024 -0.178 -0.017 -0.008
f(IDEOLOGY)2  -0.140  0.031 -0.032  0.050  0.161  0.153  0.116  0.125 -0.021
f(IDEOLOGY)3  -0.098  0.013 -0.012  0.030 -0.017 -0.037 -0.042 -0.019 -0.101
f(AFR_HISTW    -0.462 -0.056 -0.163  0.114 -0.071 -0.075 -0.068 -0.078 -0.020
              f(IDEOLOGY)2 f(IDEOLOGY)3

```

```

DELTA_2_LIB
logGNI
logPOPULATN
REGIONCntrl
REGIONEast
REGIONSouth
REGIONWest
COL2NotClny
f(IDEOLOGY)2
f(IDEOLOGY)3  0.113
f(AFR_HISTW   0.093      0.129
optimizer (nloptwrap) convergence code: 0 (OK)
boundary (singular) fit: see help('isSingular')

```

Random effects estimates

X.Intercept.

1965	-0.63668968
1975	-0.23825000
1985	-0.06810880
1995	0.03110677
2000	0.14980935
2005	0.21119287
2010	0.23339795
2015	0.33648688